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Volatility connectedness of commodity futures and its application in portfolio optimization

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This paper investigates volatility connectedness and its application in portfolio optimization, focusing on China's commodity futures market. Using the Time-Varying Parameter Vector Autoregression model, we examine the dynamic connectedness of commodities. In addition to realized volatility, continuous volatility is introduced to reduce distortions caused by intraday jumps. We develop new strategies by (1) incorporating connectedness penalties into the objective function and (2) imposing additional constraints on commodity weights based on net connectedness and degree centrality. These strategies are compared to traditional Markowitz and other benchmark portfolios. Our findings demonstrate that the proposed strategies significantly outperform the benchmark portfolios, particularly during crisis periods. These results are robust to alternative volatility measures, different lag lengths, extended forecast horizons, and various adjustments to key model parameters. Furthermore, the continuous volatility-based strategies outperform their realized volatility counterparts in most cases, offering enhanced stability and reduced short positions. These findings offer valuable insights into the role of connectedness in mitigating systemic risks and optimizing portfolio strategies, emphasizing the necessity of incorporating both covariance and volatility connectedness in the decision-making process.

Keywords: Volatility connectedness; Commodity futures; Portfolio optimization

1. Introduction

Commodities, as fundamental raw materials, play a crucial role in economic development. Over the past three decades, the commodity market has experienced significant volatility, making it one of the most volatile asset classes (Chiang *et al.* 2015, Kang *et al.* 2020). Despite being a substantial source of risk, commodities have been empirically shown to enhance portfolio performance when added to traditional equity and bond portfolios (Rad *et al.* 2022). Recent academic research on commodity markets covers topics like financialization (Tang and Xiong 2012, Henderson *et al.* 2015; Basak and Pavlova 2016), hedging strategies (Chng 2009, Acharya *et al.* 2013, Abid *et al.* 2020), pricing kernel (Christoffersen *et al.* 2013, Linn *et al.* 2018), and systemic risks (Ouyang *et al.* 2022, 2024, Wang 2023).

Among the many topics in the commodity market, risk spillover is of significant academic interest. Various methods have been employed to study risk spillover, including Granger causality, minimum spanning trees (MST), and related approaches (e.g., Billio *et al.* 2012; Wang *et al.* 2021; Huang *et al.* 2021; Shi *et al.* 2022). Most notably, Diebold and Yilmaz (2012) introduced a connectedness measure to detect risk spillover, with the network represented by an adjacency matrix (variance decomposition matrix) that shows the share of forecast error variance attributed to shocks from different sources. Motivated by their seminal study, several scholars have extended the connectedness framework. For example, Baruník and Křehlík (2018) introduce a frequency-domain decomposition method to analyze connectedness at various horizons. Ando *et al.* (2022) develop a new connectedness measure based on quantile VAR model in order to distinguish different volatility ranges. Demirer *et al.* (2018) integrate the Diebold-Yilmaz methodology with a LASSO-VAR model to construct high-dimensional networks, thereby enhancing its applicability. Considering that the connectedness approach can reveal asymmetric influences and incorporate ex-ante information from VAR models, we adopt this method to examine risk spillovers among commodity markets in this paper.

Volatility measures, such as intraday and implied volatility, have been widely used to document spillover effects (Ahmad *et al.* 2021, Ameur *et al.* 2021, Di and Xu 2022). However, realized volatility typically exhibits 'jumpy' behavior, reflecting infrequent but large market movements (Duong

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and Swanson 2015, Gkillas *et al.* 2018, Huang *et al.* 2022). To obtain a more reliable measure of underlying volatility dynamics, it is useful to distinguish continuous volatility from jump components within realized volatility. Specifically, continuous volatility provides a smoother and more stable representation of volatility due to the exclusion of sporadic jumps, thereby reducing estimation errors in the TVP-VAR framework and delivering a more consistent indicator of systemic risk. In contrast, jump volatility primarily captures transient market shocks rather than sustained volatility spillovers. Prior studies suggest that jumps may distort volatility spillover analyses and reduce portfolio diversification effectiveness (Liu and Tu 2012). Therefore, in our study, both realized volatility and continuous volatility are included to explicitly examine and compare their relative performance, where continuous volatility is considered due to its potential advantages in stability, estimation accuracy, and capturing persistent volatility dynamics relevant to systemic risk assessment. Various estimators, such as the realized bi-power variation (Barndorff-Nielsen and Shephard 2006), MinRV and MedRV (Andersen *et al.* 2012), and some threshold techniques including CPR test (Corsi *et al.* 2010) and PZ tests (Podolskij and Ziggel 2010), have been developed. In this study, we use the MinRV estimator (Andersen *et al.* 2012) to extract jumps and define continuous volatility as the difference between realized volatility and significant jumps for studying volatility connectedness.

China's commodity market holds economic significance, accounting for 72.3% of global trading volume in 2022. The market's increasing openness, as evidenced by the expansion of foreign-accessible commodity futures from 9 to 41 in 2022, attracts global investors. This study focuses on the commodity futures market, as futures contracts are standardized and actively traded on exchanges, unlike the dispersed spot market. To investigate the dynamics of risk spillover in China's commodity market, we analyze the connectedness of both realized and continuous volatility using the Time-Varying Parameter Vector Autoregression (TVP-VAR) model. The analysis suggests that continuous volatility connectedness is more sensitive to extreme events and captures changes in systemic risks better than realized volatility.

Volatility connectedness provides valuable insights for portfolio construction. Although Hoang and Baur (2021) argue that spillover information is inherently embedded in covariance matrices and thus redundant for asset allocation, we contend that volatility connectedness offers incremental and distinct advantages. Unlike covariance matrices, which are based on pairwise relationships and assume symmetry, spillover measures are derived from a multivariate framework and variance decomposition approach within a vector autoregression model. As a result, spillover matrices inherently capture directional and asymmetric aspects of risk transmission, highlighting how risk contributions flow among assets within the forecast horizon.

Traditional portfolio optimization approaches, such as the Markowitz model, rely predominantly on covariance matrices to model asset dependencies. Despite their theoretical simplicity and elegance, these methods face several practical challenges. For instance, Markowitz optimization frequently produces extreme negative weights (Green and Hollifield 1992), suffers from error maximization problems (Michaud, 1989), and demonstrates high sensitivity to changes in estimated means and variances (Best and Grauer 1991, Chopra 1993). Moreover, conventional covariance-based methods often assume static, symmetric interdependencies, potentially failing to fully capture the dynamic, directional, and asymmetric nature of risk transmission, especially during periods of market turmoil and systemic stress.

In contrast, our proposed approach leverages a TVP-VAR framework, combined with multivariate Kalman filtering. This method offers a more flexible and robust estimation of dynamic volatility structures, thus addressing the robustness issues and severe estimation errors commonly associated with traditional Markowitz optimization. Furthermore, the TVP-VAR approach naturally facilitates volatility connectedness analyses, producing a network-based representation of volatility spillovers. Such networks reveal the evolving nature of systemic risk, clearly distinguishing risk senders and receivers, quantifying the strength of directional risk transmission, and capturing structural shifts in market interconnectedness over time.

By incorporating dynamic volatility connectedness into portfolio construction, investors can develop more informed and adaptive strategies that better recognize and respond to latent systemic risks and evolving market conditions. Therefore, we develop new portfolio strategies based on volatility connectedness and evaluate their performance. The Markowitz model can be modified by adjusting the objective functions (Tola *et al.* 2008, Tsao 2010, Pantaleo *et al.* 2011, Shao *et al.* 2016) or imposing additional constraints (Jobst *et al.* 2001, Deng *et al.* 2012, Martin and Swanson 2022). For objective function modification, we propose the 'PCI-Adjusted' and 'NC-Adjusted' strategies, incorporating penalties based on the PCI (Pairwise Connectedness Index) and NC (net connectedness). These penalties are applied by either subtracting them from the Sharpe ratio or adjusting the covariance matrix. Following Pedersen *et al.* (2021) and Torrente and Uberti (2021), we treat NC as a 'negative attractiveness', subtracting the weighted-average NC from the Sharpe ratio. Motivated by Broadstock *et al.* (2022), who originally proposed the minimum connectedness portfolio, and subsequently applied by Pham and Do (2022), we incorporate PCI into the covariance matrix, thereby enlarging the denominator of the Sharpe ratio and reducing its value.

Unlike Broadstock *et al.* (2022), Pham and Do (2022) and Bai *et al.* (2023), we retain the covariance matrix while incorporating spillover information. For constraint modification, we introduce the 'NC (Net Connectedness)-Restricted' and 'DC (Degree Centrality)-Restricted' strategies, building on Peralta and Zareei (2016) and Výrost *et al.* (2019), who demonstrated a negative relationship between asset centrality and optimal weights, as well as the role of volatility connectedness in enhancing asset selection. The portfolio performance is evaluated via accumulated returns, maximum drawdown, Sharpe ratio, and winning ratio. Our findings show that the proposed strategies significantly outperform the benchmark portfolios.

Our contribution to the literature is threefold. First, we extend the application of connectedness networks to portfolio strategy, moving beyond the traditional focus on network characteristics such as spillover magnitude, direction, and clustering (Guo *et al.* 2021, Liu *et al.* 2022, Wu *et al.* 2022). While some studies examine portfolio implications (Zhang *et al.* 2021, Alomari *et al.* 2022, Trabelsi *et al.* 2022), they are typically confined to pairwise hedging ratios. In contrast, our connectedness-based strategy integrates multiple commodities, utilizing the rich information embedded in spillover networks. Furthermore, the 'PCI-Adjusted' strategy provides a generalizable framework for portfolio optimization: when the PCI-penalty term is zero, the model aligns with the Markowitz model; as the penalty increases, it converges to the MCoP model proposed by Broadstock *et al.* (2022).

Second, we offer a novel perspective on volatility spillover by focusing on continuous volatility, in contrast to previous studies that rely on intraday or implied volatility (Ahmad *et al.* 2021, Ameur *et al.* 2021, Di and Xu 2022). To our knowledge, this is the first investigation into continuous volatility connectedness and its application in enhancing investment strategies. Our findings indicate that jumps can distort spillover effects, highlighting the importance of excluding them for a clearer understanding of systemic and portfolio risks at both the macro and micro levels.

Third, our study bridges investment theory and network theory. Previous research shows that assets selected by the mean-variance portfolio tend to occupy the network periphery (Onnela *et al.* 2003) and investing in peripheral stocks can reduce portfolio risks (Pozzi *et al.* 2013). Peralta and Zareei (2016) theoretically establish a negative relationship between network centrality and optimal portfolio weights, proposing a centrality-dependent strategy within the Markowitz framework. Our findings build on this body of literature, offering new insights into the role of centrality in asset allocation.

The remainder of this paper is structured as follows: Section 2 outlines the methodology. Section 3 details the data. Section 4 analyses the dynamic connectedness of realized and continuous volatility. Section 5 presents the connectedness-based portfolio optimization. Section 6 provides robustness tests. Section 7 concludes.

2. Methodology

2.1. Minimum realized volatility

Accurate jump detection is essential for understanding volatility dynamics, and several methods have been developed for this purpose. Common approaches for extracting jumps include bi-power and tri-power variation estimators. In this paper, we adopt the minimum realized volatility (MinRV) method, introduced by Andersen *et al.* (2012), to identify jumps. The jump extraction process involves the following steps. First, we assume that the logarithmic price of commodity futures follows a standard jump diffusion process, as described by Andersen *et al.* (2011):

$$dp(\tau) = \mu(\tau)d\tau + \sigma(\tau)dw(\tau) + k(\tau)dp(\tau) \tag{1}$$

where $\tau \in R^2$, $\mu(\tau)$ represents the drift term with a continuous variation sample path; $\sigma(\tau)$ refers to the diffusion parameter. $\omega(\tau)$ stands for a standard Brownian process; $k(\tau)dp(\tau)$ is the pure jump part.

Intraday volatility can be characterized by the quadratic variation of the price process over the course of a given trading day (Barndorff-Nielsen and Shephard 2002) as below.

$$\mathbf{QV}_t = \int_{t-1}^{t} \sigma_s^2 ds + \sum_{t-1<s\leq t} k_s^2 \tag{2}$$

The integral part represents the continuous component of volatility, while the remaining portion accounts for the jumps. To calculate the continuous volatility, we subtract the jumps from the total quadratic variation. Realized variance ($\mathbf{RV}_t$) is used to estimate quadratic variation, as it converges to the true quadratic variation ($\mathbf{QV}_t$) in probability:

$$\mathbf{RV}_t = \sum_{i=1}^{N_t} r_{i,t}^2 \tag{3}$$

where $r_{i,t}$ refers to the logarithmic returns of commodity futures i on trading day t. N_t represents the number of returns observed on trading day t. Using the nearest neighbor truncation, we construct the MinRV estimator as shown in equation (4), which removes jumps by selecting the minimum value between adjacent returns.

$$\mathbf{MinRV}_t = \frac{\pi}{\pi-2}\left(\frac{N_t}{N_t-1}\right)\sum_{i=1}^{N_t-1} \mathrm{Min}(|r_{t,i}|, |r_{t,i+1}|)^2 \tag{4}$$

2.2. Connectedness index

The connectedness index, introduced by Diebold and Yilmaz (2012), is constructed using the vector autoregression (VAR) model. After testing the stationarity of the volatility series x_t, we determine the optimal lag order (p) based on the AIC criterion. The coefficient matrix ϕ_i is then estimated. To better capture the dynamic changes in the system, the coefficient matrix is transformed into its moving average form $\mathbf{A}_i$ via through iterative steps, where $\mathbf{A}_0$ is the identity matrix of order N.

$$\mathbf{x}_t = \sum_{i=1}^{p} \phi_i \mathbf{x}_{t-i} + \boldsymbol{\varepsilon}_t, \quad \varepsilon \sim N(0, \boldsymbol{\Sigma}) \tag{5}$$

$$\mathbf{x}_t = \sum_{i=0}^{\infty} \mathbf{A}_i \boldsymbol{\varepsilon}_{t-i} \tag{6}$$

$$\mathbf{A}_i = \phi_1 \mathbf{A}_{i-1} + \phi_2 \mathbf{A}_{i-2} + \cdots + \phi_p \mathbf{A}_{i-p} \tag{7}$$

We then perform variance decomposition to measure the contribution of commodity j to H-step-ahead forecast variance error of commodity i.

$$\theta_{ij}^{g}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} \left(\mathbf{e}_i' \mathbf{A}_h \sum \mathbf{e}_j\right)^2}{\sum_{h=0}^{H-1} \mathbf{e}_i' \mathbf{A}_h \sum \mathbf{A}_h' \mathbf{e}_i} \tag{8}$$

where $\sum$ represents the covariance matrix of residuals, σ_{jj} is the standard error of j^{th} residual terms, and e_i is the selection vector, which has an element of 1 for commodity i, while all other elements are zero. Since the generalized variance decomposition allows for non-orthogonal shocks, the sum of the row elements in the variance decomposition matrix does not necessarily equal 1. Therefore, normalization is required.

$$\tilde{\theta}_{ij}^{g}(H) = \frac{\theta_{ij}^{g}(H)}{\sum_{j=1}^{N} \theta_{ij}^{g}(H)} \tag{9}$$

To assess the magnitude of spillover effects within the entire network, we calculate the overall connectedness by averaging all the non-diagonal elements of the variance decomposition matrix.

$$S^{g}(H) = \frac{\sum_{i,j=1}^{N} \tilde{\theta}_{ij}^{g}(H)}{N} * 100(i \neq j) \tag{10}$$

Additionally, we compute the total directional connectedness. Equation (11) shows the 'From connectedness' which measures the cumulative shocks that commodity i receives from other commodities. Equation (12) is the 'To connectedness' which reflects the total risk that commodity i transmits to others.

$$S_{i\leftarrow \cdot}^{g}(H) = \frac{\sum_{j=1}^{N} \tilde{\theta}_{ij}^{g}(H)}{N} * 100(i \neq j) \tag{11}$$

$$S_{\cdot \leftarrow i}^{g}(H) = \frac{\sum_{j=1}^{N} \tilde{\theta}_{ji}^{g}(H)}{N} * 100(i \neq j) \tag{12}$$

To identify each commodity's role within the network, we calculate net connectedness by subtracting the 'From connectedness' from the 'To connectedness'.

$$S_{i}^{g}(H) = S_{\cdot \leftarrow i}^{g}(H) - S_{i\leftarrow \cdot}^{g}(H) \tag{13}$$

If the net connectedness is positive, the commodity is classified as a net risk transmitter; otherwise, it is classified as a net risk receiver.

Given the evolving nature and volatility of financial markets, models with single fixed parameters are inadequate for capturing the complexity over an entire sample period (Diebold and Yilmaz 2012). Therefore, we extend our analysis to dynamic spillovers using the Time-Varying Parameter Vector Autoregression (TVP-VAR) methodology introduced by Antonakakis *et al.* (2020). This approach enhances the original connectedness framework of Diebold and Yılmaz (2014) by allowing the variance-covariance matrix to evolve dynamically via a Kalman-filter estimation with forgetting factors, as proposed by Koop and Korobilis (2014). In line with Koop and Korobilis (2014), we adopt benchmark forgetting factor values of $k_1 = 0.99$ and $k_2 = 0.96$ in this study. The TVP-VAR(p) model can be written as follows:

$$\mathbf{y}_t = \mathbf{A}_t \mathbf{z}_{t-1} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t | \boldsymbol{\Omega}_{t-1} \sim \mathrm{N}(0, \boldsymbol{\Sigma}_t)$$

$$\mathrm{vec}(\mathbf{A}_t) = \mathrm{vec}(\mathbf{A}_{t-1}) + \boldsymbol{\zeta}_t, \quad \boldsymbol{\zeta}_t | \boldsymbol{\Omega}_{t-1} \sim \mathrm{N}(0, \boldsymbol{\Xi}_t)$$

$$\mathbf{z}_{t-1} = \begin{bmatrix} \mathbf{y}_{t-1} \\ \mathbf{y}_{t-2} \\ \vdots \\ \mathbf{y}_{t-p} \end{bmatrix} \mathbf{A}_\mathbf{t}' = \begin{bmatrix} \mathbf{A}_{1t} \\ \mathbf{A}_{2t} \\ \vdots \\ \mathbf{A}_{pt} \end{bmatrix}$$

where $\boldsymbol{\Omega}_{t-1}$ represents all information available up to time t-1; $\mathbf{y}_t$ and z_{t-1} represents $m \times 1$ and $mp \times 1$ vectors, respectively; $\mathbf{A}_t$ and $\mathbf{A}_{it}$ are $m \times mp$ and $m \times m$ dimensional matrices, respectively; $\boldsymbol{\varepsilon}_t$ is an $m \times 1$ vector; $\boldsymbol{\zeta}_t$ is an $m^2 p \times 1$ dimensional vector; the time-varying variance-covariance matrices $\boldsymbol{\Sigma}_t$ and $\boldsymbol{\Xi}_t$ are $m \times m$ and $m^2 p \times m^2 p$ dimensional matrices, respectively. Moreover, vec($\mathbf{A}_t$) is the vectorization of which is an $m^2 p \times 1$ dimensional vector. The multivariate Kalman filter can be formulated as follows:

$$\begin{aligned}
&\mathrm{vec}(\mathbf{A}_t)|\mathbf{z}_{1:t-1} \sim \mathrm{N}(\mathrm{vec}(\mathbf{A}_{t|t-1}), \boldsymbol{\Sigma}_{t|t-1}^{\mathbf{A}}) \\
&\mathbf{A}_{t|t-1} = \mathbf{A}_{t-1|t-1} \\
&\boldsymbol{\varepsilon}_t = \mathbf{y}_t - \mathbf{A}_{t|t-1}\mathbf{z}_{t-1} \\
&\boldsymbol{\Sigma}_t = k_2 \boldsymbol{\Sigma}_{t-1|t-1} + (1 - k_2)\boldsymbol{\varepsilon}_t' \boldsymbol{\varepsilon}_t \\
&\boldsymbol{\Xi}_t = (1 - k_1^{-1})\boldsymbol{\Sigma}_{t-1|t-1}^{\mathbf{A}} \\
&\boldsymbol{\Sigma}_{t|t-1}^{\mathbf{A}} = \boldsymbol{\Sigma}_{t-1|t-1}^{\mathbf{A}} + \boldsymbol{\Xi}_t \\
&\boldsymbol{\Sigma}_{t|t-1}^{\mathbf{A}} = \mathbf{z}_{t-1}\boldsymbol{\Sigma}_{t|t-1}^{\mathbf{A}}\mathbf{z}_{t-1}' + \boldsymbol{\Sigma}_t
\end{aligned}$$

We updated $\mathbf{A}_t$, $\boldsymbol{\Sigma}_t^{\mathbf{A}}$ and $\boldsymbol{\Sigma}_t$ based on the information available at time t through the following steps:

$$\begin{aligned}
&\mathrm{vec}(\mathbf{A}_t)|\mathbf{z}_{1:t} \sim \mathrm{N}(\mathrm{vec}(\mathbf{A}_{t|t}), \boldsymbol{\Sigma}_{t|t}^{\mathbf{A}}) \\
&\mathbf{K}_t = \boldsymbol{\Sigma}_{t|t-1}^{\mathbf{A}}\mathbf{z}_{t-1}'\boldsymbol{\Sigma}_{t|t-1}^{-1} \\
&\boldsymbol{\Sigma}_{t|t}^{\mathbf{A}} = (\mathbf{I} - \mathbf{K}_t)\boldsymbol{\Sigma}_{t|t-1}^{\mathbf{A}} \\
&\boldsymbol{\varepsilon}_{t|t} = \mathbf{y}_t - \mathbf{A}_{t|t}\mathbf{z}_{t-1} \\
&\boldsymbol{\Sigma}_{t|t} = k_2\boldsymbol{\Sigma}_{t-1|t-1} + (1 - k_2)\boldsymbol{\varepsilon}_{t|t}'\boldsymbol{\varepsilon}_{t|t}
\end{aligned}$$

where $\mathbf{K}_t$ represents the Kalman gain that determines the extent to which the parameters, $\mathbf{A}_t$, should be adjusted in each state. $\boldsymbol{\Sigma}_{t|t-1}^{\mathbf{A}}$ denotes the parameter uncertainty associated with $\mathbf{A}_t$.

3. Data

Considering availability and representativeness, we use 15-minute closing prices of the dominant futures contract† for 17 commodities sourced from the Wind database, as shown in table 1. The assets included in our sample are selected based on the composition of the China Commodity Price Index (CCPI). In addition to high-frequency data, we also extract daily opening prices and settlement prices from the Wind database. The sample period‡ spans from April 2019 to April 2022, with January 23, 2020, marked as the outbreak of COVID-19. After excluding non-trading days and handling missing values, we merge the data by dates, resulting in 721 observations for each commodity time series.

Table 2 presents the descriptive statistics of realized volatility, calculated by accumulating the squared intraday returns as shown in equation (3). The data reveals that iron ore (I)

† The dominant futures contract is defined as the most actively traded contract in the market.
‡ Since crude oil futures were listed on March 26, 2018, we lag the sample period by one year to account for the price discovery process.

Table 1. Symbol of commodity.

Commodity	Symbol	Commodity	Symbol
Crude oil	*SC*	Iron Ore	*I*
Gold	*AU*	Rubber	*RU*
Coke	*J*	Soybean	*A*
Coking Coal	*JM*	Corn	*C*
Steam Coal	*ZC*	Palm Oil	*P*
Copper	*CU*	Cotton	*CF*
Aluminum	*AL*	Sugar	*SR*
Zinc	*ZN*	Rebar	*RB*
Lead	*PB*		

Note: The symbol represents the ticker of the corresponding futures contracts traded on the exchange.

and steam coal (ZC) exhibit the highest average volatility, followed by coking coal (JM). We observe that commodities within the coal sector show higher volatility compared to those in the non-ferrous metals and agricultural products sectors. Additionally, coking coal (JM) and iron ore (I) display the most dispersed volatility, as indicated by their standard deviations. All commodities demonstrate positive skewness, with soybean (A) showing the highest positive skewness, reflecting frequent small losses and occasional extreme gains. The kurtosis for all commodities exceeds three, indicating fat tails and sharper peaks in the volatility distributions. The results of the ADF test confirm that all commodity volatilities are stationary.

Continuous volatility is typically driven by normal market conditions and provides deeper insights into market fundamentals (Huang *et al.* 2022). Therefore, we adopt the minimum realized volatility (MinRV) method and obtain the continuous volatility by filtering out intra-day jumps from the integrated variance. Table 3 presents the descriptive statistics of continuous volatility. The data clearly shows that, compared to realized volatility, the continuous component for each commodity exhibits lower mean and standard deviation values. The distribution of continuous volatility is also less skewed and less leptokurtic, indicating a reduction in tail risks after removing jumps from the overall volatility.

4. Dynamic connectedness of volatility

4.1. Overall connectedness

We aim to examine the time-varying nature of connectedness based on both realized volatility and continuous volatility, with the latter having its jump component filtered out. Figure 1 illustrates the dynamics of overall connectedness using these two different volatility measures. For comparison, both curves are presented in the same figure. The dashed line represents the net connectedness of realized volatility, while the solid line shows the net connectedness of continuous volatility.

In general, continuous volatility connectedness follows a similar trend to realized volatility connectedness. However, deviations between the two trends emerge during the highlighted periods. In the first period, Wuhan was placed under lockdown on January 23, 2020, due to COVID-19, and soon afterward, the WHO declared the virus a global pandemic. The COVID-19 pandemic has led to global economic policy uncertainty, increasing the need to explore ways to mitigate this uncertainty (Aftab *et al.* 2023; Chen *et al.* 2022; Li *et al.* 2022). As a result, we expected to see an upward trend in overall connectedness. Yet only continuous volatility connectedness exhibited this upward trend. In the second period, China's implementation of the power restriction policy, which significantly impacted industrial production, represented a supply-side shock that should have also increased systemic risks. Again, we observed a rising trend only in continuous volatility connectedness.

Therefore, we argue that continuous volatility connectedness better captures the dynamics of systemic risks compared to realized volatility connectedness. In an un-tabulated analysis, we found that continuous volatility connectedness exhibits greater magnitude on average and fluctuates more intensely than realized volatility connectedness. This suggests that volatility jumps can suppress spillover effects between commodities and further implies that the overall connectedness index based on continuous volatility may serve as a more effective proxy for systemic risks.

4.2. Net connectedness

We further examine the dynamic net connectedness of each commodity based on both realized volatility and continuous volatility. Figure 2 illustrates the dynamics of net connectedness using these two different volatility measures. For comparison purpose, both curves are presented in the same figure. The dashed line represents the net connectedness of realized volatility, while the solid line shows the net connectedness of continuous volatility. To demonstrate the roles that each commodity plays in risk transmission, we introduce a baseline of zero net connectedness.

The curves of net connectedness for each commodity fluctuate around zero, reflecting frequent role reversals throughout the sample period. Similar to the results in figure 1, we observe that net connectedness of continuous volatility shows more intense fluctuations, marked by larger spikes. Several commodities experienced significant spikes in net connectedness. For instance, the most notable jumps in iron ore (I) can be attributed to supply-side shocks, such as the mining disaster in Brazil and the hurricane in Australia in 2019. Notably, some commodities exhibited immediate role reversals within a three-day window from January 21 to January 23, 2020, coinciding with the Wuhan lockdown in response to COVID-19. For example, crude oil (SC) shifted from net risk transmitter to net risk receiver, then back to net risk transmitter. These patterns are not observed in net realized volatility connectedness, suggesting that net continuous volatility connectedness is more sensitive to extreme events, consistent with our findings on overall connectedness. For illustrative purposes, we present the net connectedness of crude oil (SC), gold (AU) and iron ore (I) in figure 2, while the net connectedness of the remaining commodities is provided in the Appendix A.

Table 2. Descriptive statistics of realized volatility.

Commodity	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis	ADF
SC	0.046	0.061	0.001	0.582	4.443	29.523	– 10.093***
AU	0.007	0.011	0.000	0.177	7.533	93.858	– 13.010***
J	0.043	0.068	0.001	0.796	5.168	40.126	– 5.756***
JM	0.049	0.085	0.001	1.119	5.300	47.387	– 6.258***
ZC	0.056	0.189	0.000	3.664	12.418	207.450	– 11.339***
CU	0.010	0.012	0.000	0.126	4.093	29.119	– 9.436***
AL	0.015	0.028	0.000	0.509	9.873	148.838	– 9.715***
ZN	0.016	0.037	0.000	0.800	15.843	304.319	– 12.812***
PB	0.012	0.014	0.000	0.267	9.730	160.381	– 12.214***
I	0.066	0.147	0.004	2.688	13.223	207.444	– 13.899***
RU	0.024	0.068	0.002	1.481	16.246	314.321	– 14.972***
A	0.013	0.038	0.000	0.917	20.254	470.898	– 15.396***
C	0.005	0.009	0.000	0.179	11.744	191.700	– 14.044***
P	0.034	0.106	0.000	1.810	13.650	210.551	– 14.789***
CF	0.014	0.029	0.000	0.419	8.517	96.566	– 12.186***
SR	0.007	0.011	0.000	0.248	15.959	34.278	– 15.043***
RB	0.020	0.035	0.000	0.605	10.124	146.031	– 11.283***
No. of observations				721			

Note: Mean, standard deviation, minimum, and maximum values are expressed in percentage terms. The ADF test statistic is used to assess stationarity. The asterisks (***) indicate rejection of the unit root hypothesis at the 1% significance level.

Table 3. Descriptive statistics of continuous volatility.

Commodity	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis	ADF
SC	0.028	0.030	0.001	0.273	3.409	17.056	– 8.135***
AU	0.004	0.005	0.000	0.050	4.404	28.906	– 9.988***
J	0.032	0.059	0.000	0.901	8.036	91.616	– 7.799***
JM	0.036	0.065	0.000	0.949	6.586	68.055	– 9.043***
ZC	0.032	0.063	0.000	0.653	4.734	31.741	– 8.291***
CU	0.007	0.008	0.000	0.097	4.442	30.953	– 10.522***
AL	0.011	0.016	0.000	0.154	4.193	23.418	– 8.426***
ZN	0.010	0.011	0.000	0.132	4.367	31.458	– 9.258***
PB	0.009	0.008	0.001	0.073	3.288	17.224	– 9.817***
I	0.046	0.045	0.001	0.423	2.816	13.114	– 8.321***
RU	0.016	0.020	0.001	0.261	5.230	43.985	– 9.550***
A	0.009	0.008	0.000	0.076	2.603	11.763	– 10.792***
C	0.004	0.003	0.000	0.028	2.821	12.465	– 9.325***
P	0.020	0.020	0.000	0.179	3.326	17.273	– 8.591***
CF	0.010	0.014	0.001	0.203	6.483	64.483	– 10.707***
SR	0.005	0.005	0.000	0.062	4.788	40.908	– 11.224***
RB	0.015	0.018	0.000	0.230	4.563	35.856	– 7.261***
No. of observations				721			

Note: Mean, standard deviation, minimum, and maximum values are expressed in percentage terms. The ADF test statistic is used to assess stationarity. The asterisks (***) indicate rejection of the unit root hypothesis at the 1% significance level.

5. Portfolio optimization with volatility connectedness

5.1. Overview of strategy design

Our aim is to apply volatility connectedness to portfolio optimization by modifying the conventional Markowitz model. One approach to improving the performance of Markowitz portfolios is by modifying the objective functions (Tola *et al.* 2008, Tsao 2010, Pantaleo *et al.* 2011, Shao *et al.* 2016). Another approach is to impose additional constraint conditions (Jobst *et al.* 2001, Deng *et al.* 2012, Martin and Swanson 2022). In these different approaches, we take the connectedness of both realized volatility and continuous volatility into account.

For strategy design, we use a rolling-window method, starting from February 21, 2020, the 201st day in our sample period. The formation period spans the previous 200 trading days. Short selling of commodities is allowed, restricting weights to $[-1, 1]$.† Positions are entered at the open price

† We impose symmetric portfolio weight constraints of $[-1, 1]$ for each futures contract to ensure economic realism and practical implementability. This approach aligns closely with common risk management practices in the China's futures market (e.g., margin requirements, position limits, and broker-level risk controls), helps

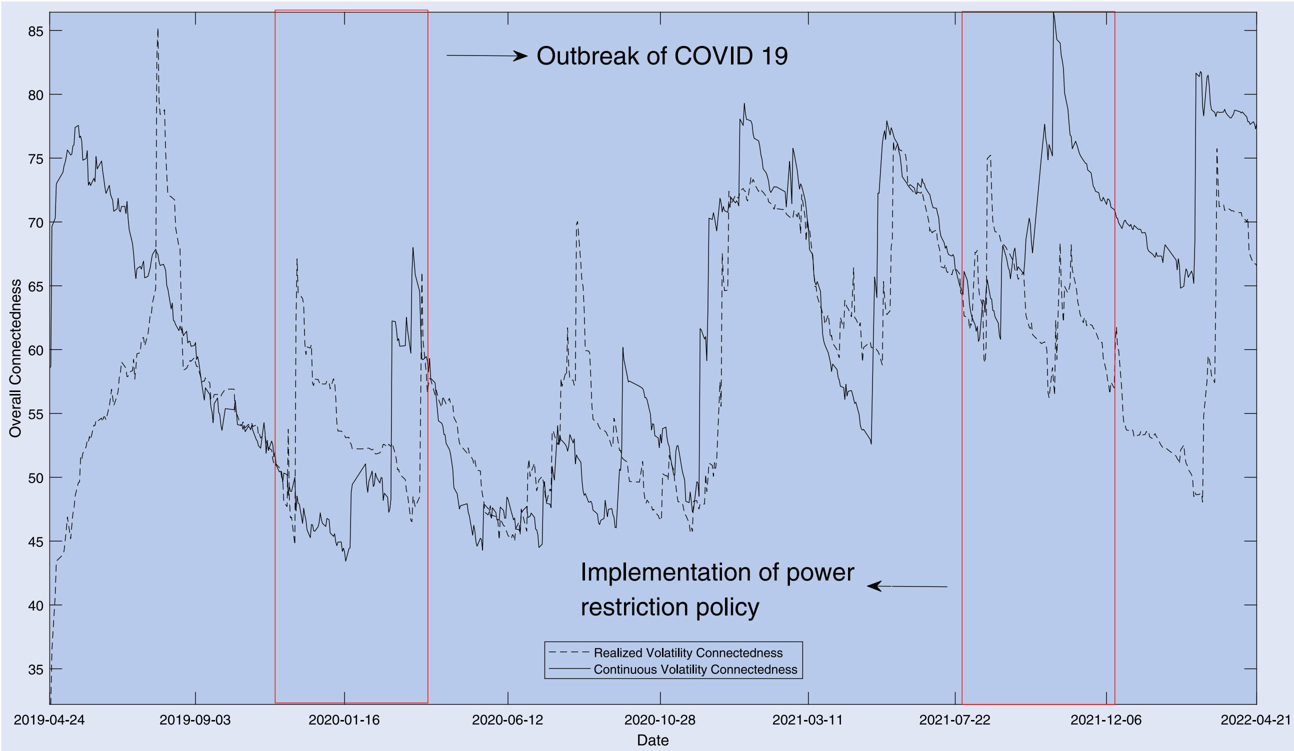


Figure 1. Overall connectedness: realized volatility and continuous volatility.
Note: This figure illustrates the dynamic overall connectedness of realized and continuous volatility, estimated using the TVP-VAR method. The vertical axis denotes overall connectedness, and the horizontal axis indicates the date. The dashed line represents the overall connectedness of realized volatility, while the solid line shows the overall connectedness of continuous volatility. Two subperiods are highlighted with red rectangles: (i) the outbreak of COVID-19 and (ii) the implementation of the power restriction policy. The parameters are set as follows: lag order $p = 2$; forecast horizon $H = 10$; forgetting factor $(k_1, k_2) = (0.99, 0.96)$.

and closed at the settlement price. Portfolios are rebalanced every d days, with d = 5 for weekly trading and d = 20 for monthly trading. To evaluate portfolio performance, we use four indicators: accumulated returns, maximum drawdown, Sharpe ratio, and winning ratio. Maximum drawdown, representing the maximum potential loss, captures the tail risks of a strategy. The winning ratio is calculated as the number of transactions with positive returns divided by the total number of transactions.

5.2. Modified objective function

We propose a new investment strategy by revising the objective function of the Markowitz model. Building on the work of Broadstock *et al.* (2022), who introduced the minimum connectedness approach by substituting the covariance matrix with the PCI (Pairwise Connectedness Index), Bai *et al.* (2023) later expanded on this idea with the NPDC (Net Pairwise Directional Connectedness) index to capture portfolio risks. PCI and NPDC index are constructed as follows:

$$PCI^{g}_{ij,t}(H) = 2\left(\frac{\tilde{\theta}^{g}_{ij,t}(H) + \tilde{\theta}^{g}_{ij,t}(H)}{\tilde{\theta}^{g}_{ii,t}(H) + \tilde{\theta}^{g}_{ij,t}(H) + \tilde{\theta}^{g}_{ji,t}(H) + \tilde{\theta}^{g}_{jj,t}(H)}\right) \tag{14}$$

$$NPDC^{g}_{j\to i,t}(H) = \left(\frac{\tilde{\theta}^{g}_{ij,t}(H) - \tilde{\theta}^{g}_{ji,t}(H)}{\sum_{j=1,i=1}^{n}\tilde{\theta}^{g}_{ij,t}(H)}\right) * 100 \tag{15}$$

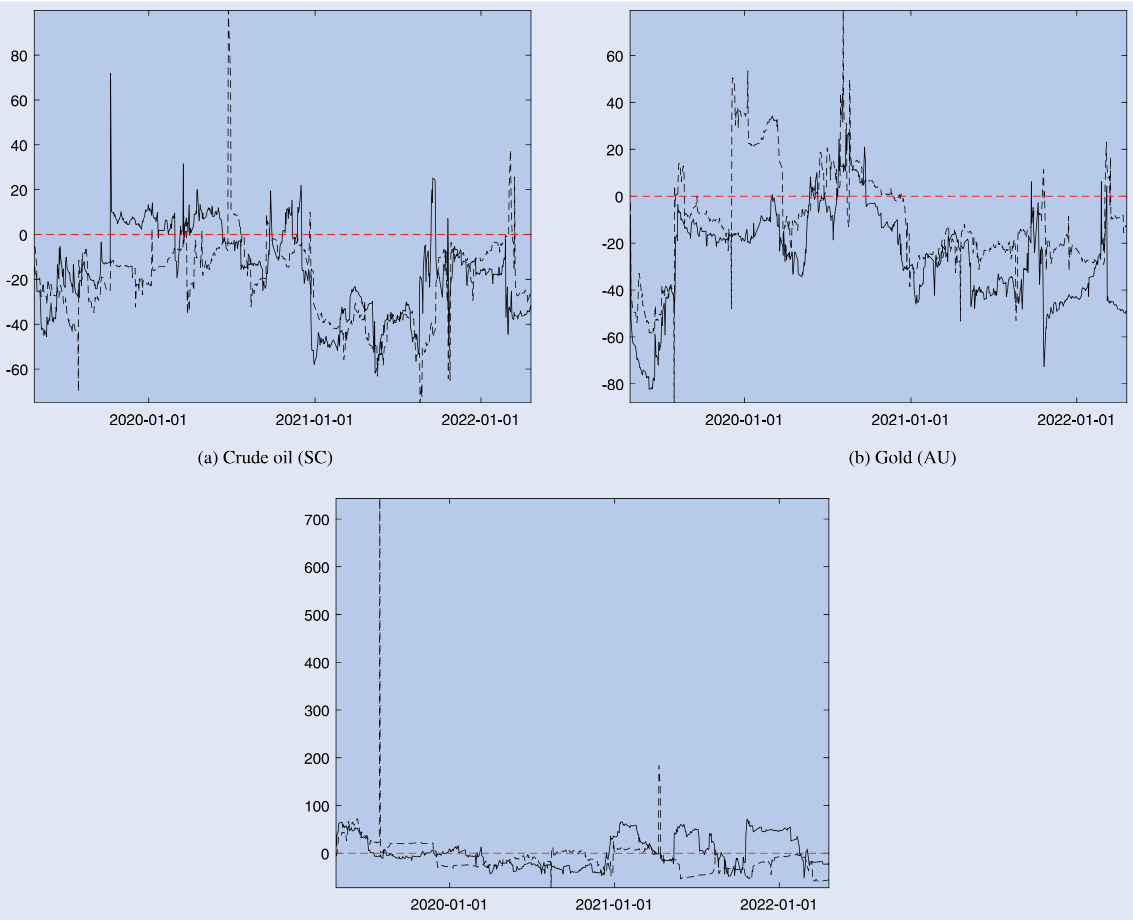
Motivated by Broadstock *et al.* (2022) and Pham and Do (2022), we design a ‘PCI-Adjusted’ strategy by incorporating a PCI-penalty term into the covariance matrix. This addresses the concern that covariance matrix estimation is often associated with statistical uncertainty. The revised objective function is shown below:

$$\max_{-1\le w_i\le 1} \frac{\mathbf{w}'\mathbf{r}}{\sqrt{\mathbf{w}'(\mathbf{C} + \alpha{*}\mathbf{PCI})\mathbf{w}}} \tag{16}$$

where $\mathbf{w}$ is a 17×1 vector of commodity weights, $\mathbf{w} = [w_1, \cdots, w_i, \cdots, w_{17}]'$; $\mathbf{r}$ is a 17×1 vector for commodity returns, $\mathbf{r} = [r_1, \cdots, r_i, \cdots, r_{17}]'$; $\mathbf{C}$ is a 17×17 covariance matrix and α is a scaling parameter set to 10^{-5} to align the order of magnitude of $\mathbf{C}$ with that of $\mathbf{PCI}$.

Since PCI is a positive and symmetric matrix, a higher PCI increases the denominator of the objective function, lowering the Sharpe ratio. In this way, greater connectedness is penalized in the portfolio’s performance. Notably, when the PCI-penalty term is zero, the PCI-adjusted model reverts to the traditional Markowitz model; when the term approaches infinity, the model converges to the minimum connectedness method. Thus, our approach offers a more generalized framework than conventional optimization models.

mitigate estimation error-driven instability commonly found in high-dimensional covariance matrices (Fan *et al.* 2012), and contributes to greater stability and robustness of portfolio performance (Behr *et al.* 2013).



(a) Crude oil (SC)

(b) Gold (AU)

(c) Iron ore (I)

Figure 2. Net connectedness: realized volatility and continuous volatility.
Note: This figure illustrates the dynamic net connectedness of realized and continuous volatility, estimated using the TVP-VAR method. The vertical axis denotes net connectedness, and the horizontal axis indicates the date. The dashed line represents the net connectedness of realized volatility, while the solid line shows the net connectedness of continuous volatility. The parameters are set as follows: lag order $p = 2$; forecast horizon $H = 0$; forgetting factor $(k_1, k_2) = (0.99, 0.96)$. (a) Crude oil (SC) (b) Gold (AU) (c) Iron ore (I).

In addition to the 'PCI-Adjusted' strategy, we propose an 'NC (Net Connectedness)-adjusted' strategy, inspired by Pedersen *et al.* (2021). They argue that some investors, motivated by ESG scores, derive positive utility from the weighted-average ESG score of the portfolio. Similarly, we posit that certain investors may care about volatility connectedness, deriving negative utility from the weighted-average net connectedness of the portfolio. This assumption is supported by Torrente and Uberti (2021), who found that connectedness negatively correlates with diversification benefits. The revised objective function for the 'NC-Adjusted' strategy is as follows

$$\max_{-1 \le w_i \le 1} \frac{\mathbf{w}'\mathbf{r}}{\sqrt{\mathbf{w}'\mathbf{C}\mathbf{w}}} - \beta * \mathbf{w}' * abs(\mathbf{NC}) \tag{17}$$

where $\mathbf{NC}$ is a 17×1 vector of net connectedness, $\mathbf{NC} = [NC_1, \cdots, NC_i, \cdots, NC_{17}]'$; β is a scaling parameter set to 10, to align the order of magnitude of the Sharpe ratio with that of $\mathbf{NC}$. We take the absolute value of $\mathbf{NC}$, as investors typically dislike assets that are closely correlated with others through connectedness, regardless of the direction.

To mitigate the impact of short-term price shocks, we first compute the average connectedness matrix over the entire formation period. We then construct the Pairwise Connectedness Index (PCI) and Net Connectedness (NC) based on this average matrix following Broadstock *et al.* (2022) and Diebold and Yılmaz (2014). The optimization problem is solved via equations (16) and (17) to obtain the portfolio weights of 'PCI-Adjusted' and 'NC-Adjusted' strategy accordingly. Table 4 presents the performance of strategy. RV and CV in brackets denote realized volatility and continuous volatility respectively, as used in the strategy. For comparison, we report the performance of two benchmark portfolios: the equal-weighted portfolio and the Markowitz portfolio. Additionally, we include MCoP1 and MCoP2 as the minimum connectedness portfolios obtained via two other strategies from the literature, i.e. the PCI strategy and the NPDC strategy, respectively.

As shown in table 4, our 'PCI-Adjusted' strategy significantly outperforms both the Markowitz and PCI strategies, with a higher Sharpe ratio, accumulated returns, winning ratio, and lower maximum drawdown. Specifically, with weekly rebalancing, the Sharpe ratio of the 'PCI-Adjusted' portfolio reaches 2.08, nearly 3.7 times that of the Markowitz portfolio (Sharpe Ratio = 0.56) and surpassing the equal-weighted portfolio (Sharpe Ratio = 1.91).

Table 4. Portfolio performance of strategy with modified objective function.

Panel A	Holding period: d = 5			
Portfolio	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	51.17%	12.61%	1.91	57.69%
Markowitz	53.23%	64.14%	0.56	51.92%
MCoP1	− 89.41%	95.34%	− 1.52	58.65%
MCoP2	16.87%	56.23%	0.19	51.92%
PCI-Adjusted (RV)	128.40%	12.36%	2.08	57.69%
PCI-Adjusted (CV)	120.66%	12.24%	2.13	57.69%
NC-Adjusted (RV)	76.38%	11.15%	2.02	54.81%
NC-Adjusted (CV)	145.10%	12.42%	2.12	57.69%
Panel B	**Holding period: d = 20**			
Portfolio	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	61.76%	9.04%	2.49	84.62%
Markowitz	46.55%	60.59%	0.46	65.38%
MCoP1	− 152.79%	119.43%	− 0.60	80.77%
MCoP2	36.60%	48.35%	0.39	53.85%
PCI-Adjusted (RV)	80.10%	7.53%	1.97	76.92%
PCI-Adjusted (CV)	82.97%	7.56%	2.04	76.92%
NC-Adjusted (RV)	50.51%	46.06%	0.55	69.23%
NC-Adjusted (CV)	77.40%	33.93%	0.88	69.23%

Note: ACC_RET refers to the accumulated returns, while MAX_DD represents the maximum drawdown. The Sharpe Ratio is calculated as the excess weekly (or monthly) returns divided by the standard deviation of returns. The Winning Ratio is the number of positive weekly (or monthly) returns divided by the total number of weekly (or monthly) returns. EW refers to the equal-weighted portfolio, while Markowitz denotes the tangency portfolio that maximizes the Sharpe ratio. MCoP1 and MCoP2 represent the minimum connectedness portfolios obtained through the PCI and NPDC strategies, respectively. PCI-Adjusted and NC-Adjusted refer to the portfolios obtained through the 'PCI-Adjusted' and 'NC-Adjusted' strategies, respectively. RV and CV, in brackets, denote realized volatility and continuous volatility, respectively.

Accumulated returns also more than double after introducing the PCI-penalty term into the Markowitz model (128.40% vs. 53.23%). Furthermore, maximum drawdown is reduced from 64.14% to 12.36%, indicating that the 'PCI-Adjusted' strategy better controls tail risks, which is crucial for risk-averse investors. After filtering out jumps, the Sharpe ratio of the 'PCI-Adjusted' portfolio improves further, and accumulated returns increase when the portfolio is rebalanced monthly. While the 'PCI-Adjusted' strategy enhances the performance of the benchmark model, the PCI and NPDC strategies represented by MCoP1 and MCoP2 accordingly, fail to improve investment outcomes. This is likely because these two naïve strategies discard much of the valuable information embedded in the covariance matrix, which is essential for effective risk management.

The results show that the 'NC-Adjusted' strategy outperforms the Markowitz portfolio across all metrics, with a higher Sharpe ratio, accumulated returns, winning ratio, and lower maximum drawdown. Its investment performance further improves when the portfolio is rebalanced weekly ($d = 5$). After excluding jumps, the 'NC-Adjusted' strategy shows improved performance in both Sharpe ratio and accumulated returns. While it surpasses the PCI and NPDC strategies, it performs slightly below the 'PCI-Adjusted' strategy.

In summary, both the 'PCI-Adjusted' and 'NC-Adjusted' strategies outperform the Markowitz model as well as the PCI and NPDC strategies. This suggests that incorporating both covariance and volatility connectedness information is essential when constructing optimal portfolios.

5.3. *Modified constraint condition*

Peralta and Zareei (2016) theoretically demonstrate a negative relationship between asset centrality in a network and their optimal weights under the Markowitz framework. To test this theory, Výrost *et al.* (2019) extend Markowitz portfolio strategies by introducing additional restrictions that assign lower weights to assets with higher centralities, improving key portfolio return characteristics. Wang *et al.* (2024) compare portfolio performance using different centrality measures and find that degree centrality is more suitable for larger portfolios. However, their study focuses solely on undirected graphs. This inspires us to introduce 'low-centrality constraints' to volatility connectedness networks, which are classified as directed graphs. Before incorporating topological information into Markowitz portfolio strategies, it is logical to first constrain commodity weights directly based on net connectedness. Therefore, we propose two strategies: the 'NC (Net Connectedness)-Restricted' strategy, based on net connectedness, and the 'DC (Degree Centrality)-Restricted' strategy, based on degree centrality. Degree centrality is defined as the number of a node's edges connecting it to other nodes, with each edge weighted by the volatility connectedness between the pairwise nodes. The new constraints for the 'NC-Restricted' and 'DC-Restricted' strategies are detailed in equations (18) and (19), respectively.

$$\mathbf{w}_i \le \mathbf{w}_j, \quad \text{if } \mathbf{NC}_i \ge \mathbf{NC}_j, \quad i \ne j \tag{18}$$

$$\mathbf{w}_i \le \mathbf{w}_j, \quad \text{if } \mathbf{DC}_i \ge \mathbf{DC}_j, \quad i \ne j \tag{19}$$

where $\mathbf{w}_i$ denotes the portfolio weights of commodity i; $\mathbf{NC}_i$ refers to the net connectedness of commodity i; $\mathbf{DC}_i$ represents the degree centrality of commodity i.

To mitigate the influence of short-term price shocks, we first compute the average connectedness matrix over the entire formation period. We then construct the Net Connectedness (NC) and Degree Centrality (DC) from this matrix. Finally, we solve the optimization problem, as outlined in equations (18) and (19), to obtain the portfolio weights. Table 5 presents the portfolio performance results for the 'NC-Restricted' and 'DC-Restricted' strategies. RV and CV in brackets denote realized volatility and continuous volatility respectively, as used in the strategy.

The results show that the 'NC-Restricted (RV)' strategy outperforms the Markowitz portfolio, with a higher Sharpe ratio, accumulated returns, winning ratio, and lower maximum drawdown. Furthermore, at a monthly rebalancing frequency, the 'NC-Restricted (CV)' strategy outperforms the 'NC-Restricted (RV)' strategy in all metrics. Regarding the 'DC-Restricted' strategy, the 'DC-Restricted (RV)' strategy outperforms the Markowitz portfolio in terms of Sharpe ratio, winning ratio, and maximum drawdown. Moreover, the 'DC-Restricted (CV)' strategy significantly surpasses 'DC-Restricted (RV)' across all performance metrics, particularly in maximum drawdown, Sharpe ratio, and winning ratio. These findings are robust across different holding periods. Therefore, we conclude that applying net connectedness and degree centrality constraints enhances the investment performance of the Markowitz strategy, with improvements more pronounced when filtering out jumps, especially for the 'DC-Restricted' strategy.

Comparing the 'NC-Restricted' and 'DC-Restricted' strategies, we find that the former generally exhibits a higher Sharpe ratio and greater accumulated returns, while the latter results in a lower maximum drawdown. Furthermore, when both strategies are implemented using continuous volatility, the maximum drawdown decreases across all strategies and holding periods, indicating that excluding jumps helps mitigate tail risks.

5.4. *Further analysis*

In summary, our connectedness-based strategy consistently outperforms the benchmark portfolio, whether by revising the objective functions or imposing additional constraints. The first approach tends to yield larger returns and higher Sharpe ratios, while the second approach achieves a higher winning ratio and mitigates maximum drawdown. This suggests that the two approaches emphasize different aspects of investment performance. Furthermore, when the strategy is based on continuous volatility (CV) rather than realized volatility (RV), the Sharpe ratio improves even further.

Notably, we observe more significant improvements in portfolio performance when filtering out jumps in the second approach. This may be because, while continuous volatility has been proved to capture systemic risks better, introducing it into the objective function removes the impact of intraday jumps, which are an integral part of market dynamics. This filtering may inadvertently exclude information that is valuable to investors, as intraday jumps often reflect short-term shocks and investor sentiment, which can influence decision-making and utility. By eliminating these jumps, the model may fail to capture certain aspects of risk that investors consider important, leading to suboptimal performance. In contrast, applying continuous volatility as an additional constraint in the second approach does not directly alter the utility framework or the fundamental risk-return trade-off. Instead, it enhances the strategy by controlling risk through constraints without assuming that jumps must be excluded from the objective function. This approach allows the model to retain the full scope of market risk while still benefiting from the filtering of extreme volatility in the constraint-setting process. As a result, it leverages the advantages of continuous volatility while avoiding the adverse effects of removing jumps from the optimization objective.

To better understand why the continuous volatility network provides a more favorable environment for investment strategies, we examine the portfolio weights of the 'DC-Restricted (RV)' and 'DC-Restricted (CV)' strategies for illustration. Table 6 presents the time-series mean and standard deviation of portfolio weights for each commodity. Two key findings emerge. First, the 'DC-Restricted (CV)' strategy reduces short positions compared to the 'DC-Restricted (RV)' strategy: (1) The number of commodities with negative weights drops from four to three; (2) Commodities like Coke (J) and Steam Coal (ZC) have smaller negative weights in the 'DC-Restricted (CV)' strategy, while Aluminum (AL) and Zinc (ZN) experience a dramatic shift in trading direction. Given that our investment period coincides with a bull market in commodities, this reduction in short positions contributes to improved portfolio performance. Second, the standard deviation of all commodity weights decreases in the 'DC-Restricted (CV)' strategy, except for Coking Coal (JM). This stability allows investors to avoid significant portfolio adjustments, indicating that filtering out jumps from realized volatility enhances the stability of the investment strategy. Overall, the 'DC-Restricted (CV)' strategy reduces exposure to short positions and smooths portfolio rebalancing, contributing to its higher Sharpe ratio and lower maximum drawdown compared to the 'DC-Restricted (RV)' strategy.

6. Robustness test

We conduct a series of robustness tests to verify the stability and effectiveness of our proposed volatility-connectedness-based portfolio strategies. These include subsample analyses, bootstrap significance tests, alternative volatility measures, and sensitivity analyses of key parameters, i.e. forgetting factors, lag lengths, forecast horizon and penalty term. The results consistently supporting the robustness and reliability of our main findings.

Table 5. Portfolio performance of strategy with modified constraints.

Panel A	Holding period: d = 5			
Portfolio	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	51.17%	12.61%	1.91	57.69%
Markowitz	53.23%	64.14%	0.56	51.92%
NC-Restricted (RV)	57.36%	18.71%	1.32	59.62%
NC-Restricted (CV)	37.67%	12.90%	1.16	56.73%
DC-Restricted (RV)	43.56%	15.67%	0.97	58.65%
DC-Restricted (CV)	39.96%	8.70%	1.69	60.58%
Panel B	Holding period: d = 20			
Portfolio	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	61.76%	9.04%	2.49	84.62%
Markowitz	46.55%	60.59%	0.46	65.38%
NC-Restricted (RV)	65.47%	17.47%	1.61	80.77%
NC-Restricted (CV)	83.02%	8.41%	2.00	76.92%
DC-Restricted (RV)	23.43%	7.56%	0.88	65.38%
DC-Restricted (CV)	49.55%	5.69%	1.93	80.77%

Note: ACC_RET refers to the accumulated returns, while MAX_DD represents the maximum drawdown. The Sharpe Ratio is calculated as the excess weekly (or monthly) returns divided by the standard deviation of returns. The Winning Ratio is the number of positive weekly (or monthly) returns divided by the total number of weekly (or monthly) returns. EW refers to the equal-weighted portfolio, while Markowitz denotes the tangency portfolio that maximizes the Sharpe ratio. NC-Restricted and DC-Restricted denote to the portfolios obtained through the 'NC-Restricted' and 'DC-Restricted' strategies, respectively. RV and CV, in brackets, denote realized volatility and continuous volatility, respectively.

Table 6. Portfolio weights with 'DC-restricted' strategy.

	Panel A Holding period: $d = 5$				Panel B Holding period: $d = 20$			
	DC-Restricted (RV)		DC-Restricted (CV)		DC-Restricted (RV)		DC-Restricted (CV)	
	Mean	Std Dev	Mean	Std Dev	Mean	Std Dev	Mean	Std Dev
SC	13.89%	12.99%	9.55%	9.20%	14.36%	14.02%	9.35%	6.83%
AU	11.50%	12.52%	9.55%	9.20%	12.06%	10.80%	9.35%	6.83%
J	− 17.24%	35.73%	− 4.41%	24.03%	− 20.60%	40.47%	− 5.52%	24.08%
JM	11.15%	12.94%	− 4.41%	24.03%	11.35%	13.80%	− 5.52%	24.08%
ZC	− 16.37%	41.60%	− 12.19%	36.73%	− 18.53%	44.32%	− 15.03%	40.36%
CU	7.94%	10.29%	8.69%	6.25%	8.02%	8.92%	9.35%	6.83%
AL	− 11.21%	36.19%	5.01%	12.56%	− 11.21%	38.69%	5.65%	13.05%
ZN	− 10.48%	32.27%	7.97%	7.06%	− 6.81%	30.03%	8.63%	6.96%
PB	7.94%	10.29%	9.45%	8.94%	8.02%	8.92%	9.35%	6.83%
I	16.20%	23.57%	8.69%	6.25%	15.82%	20.81%	9.35%	6.83%
RU	13.45%	17.57%	8.57%	6.21%	14.75%	19.06%	9.19%	6.68%
A	16.79%	15.60%	8.69%	6.25%	16.46%	16.14%	9.35%	6.83%
C	11.25%	12.91%	8.69%	6.25%	11.46%	13.76%	9.35%	6.83%
P	12.90%	15.05%	8.69%	6.25%	10.66%	10.47%	9.35%	6.83%
CF	11.15%	12.94%	9.55%	9.20%	11.35%	13.80%	9.35%	6.83%
SR	7.96%	10.29%	9.45%	8.94%	8.09%	8.90%	9.35%	6.83%
RB	13.19%	16.91%	8.69%	6.25%	14.82%	19.03%	9.35%	6.83%

Note: This table presents the time-series average (Mean) and standard deviation (Std Dev) of commodity weights under the 'DC-Restricted (RV)' and 'DC-Restricted (CV)' strategies for different holding periods. The first column lists the commodity symbols, with corresponding interpretations provided in Table 1. 'DC-Restricted (RV)' refers to the 'DC-Restricted' strategy based on realized volatility connectedness, while 'DC-Restricted (CV)' refers to the 'DC-Restricted' strategy based on continuous volatility connectedness.

6.1. Subsample analysis

Since volatility connectedness measures systemic risk and often spikes during extreme events, we examine the performance of our connectedness-based strategies across different market conditions. Specifically, we compare performance during a non-crisis period (March 2018–December 2019) and a crisis period (January 2020–April 2022). We begin the non-crisis period on March 26, 2018, namely the launch date of crude oil futures, ensuring consistency with our prior analysis of 17 commodities. The crisis period starts from January 2020, coinciding with the COVID-19 pandemic outbreak.

Tables 7 and 8 summarize strategy performance in non-crisis and crisis phases, respectively. During the non-crisis period, our strategies generally underperform the Markowitz benchmark; however, the continuous volatility connectedness-based strategies remain competitive. For example, the 'DC-Restricted' strategy outperforms the Markowitz benchmark across both rebalancing frequencies, displaying a lower maximum drawdown, higher Sharpe ratio, and superior winning ratio. This indicates that continuous volatility connectedness effectively captures asset co-movements even when risk contagion is subdued. In contrast, our strategies consistently outperform Markowitz during the crisis period, as shown in table 8. Although the 'NC-adjusted' strategy based on realized volatility underperforms, it significantly improves once jumps are filtered out, achieving higher accumulated returns, Sharpe ratios, and reduced drawdowns relative to Markowitz. Notably, continuous volatility connectedness yields substantial improvements over realized volatility, with accumulated returns rising from 49.92% to 137.95% under weekly rebalancing.

It is also worth noting that the equal-weighted (EW) portfolio frequently delivers strong performance, surpassing both the Markowitz benchmark and our connectedness-based strategies. This phenomenon, consistent with prior studies (Michaud, 1989, Best and Grauer 1991, DeMiguel *et al.* 2009), reflects the EW portfolio's robustness against estimation errors and parameter uncertainty inherent in mean-variance optimization. Therefore, it is not surprising that our optimized strategies occasionally underperform the EW portfolio. However, as our connectedness-based methods specifically address known limitations of traditional Markowitz optimization, our primary benchmark remains the Markowitz portfolio. The key advantage of our approach lies in its consistent improvement over the Markowitz framework, particularly during episodes characterized by elevated systemic risk and volatility spillovers.

In summary, our volatility-connectedness strategies exhibit greater value during crises by effectively capture intensified cross-asset spillovers and shifts in risk transmission roles, features not fully captured by conventional covariance matrices. This aligns with findings by Rubbaniy *et al.* (2024), who report similar performance gains for minimum-connectedness portfolios during market stress periods such as COVID-19 and the Russia-Ukraine conflict. Furthermore, strategies based on continuous volatility connectedness outperform their realized volatility counterparts and demonstrate greater stability across varying market conditions. This corroborates Huang *et al.* (2022), who highlight the advantage of continuous volatility in reflecting fundamental market conditions and persistent risk dynamics.

6.2. *Bootstrap significance*

To further validate the robustness of our performance comparisons, we conduct bootstrap-based analyses for selected strategies, focusing on two key dimensions: (1) testing whether our strategies yield significantly positive Sharpe ratios, and (2) assessing whether their outperformance relative to the Markowitz benchmark is statistically significant. Tables 9 and 10 report these bootstrap results for the full sample and crisis periods, respectively. For the full sample, most of our strategies exhibit significantly positive Sharpe ratios and statistically significant outperformance compared to the benchmark, particularly at the monthly rebalancing frequency. During the crisis period, these results are even stronger, as our strategies consistently deliver significantly positive Sharpe ratios and significant outperformance relative to Markowitz across all rebalancing frequencies.

6.3. *Alternative volatility measure*

To assess the impact of microstructure noise on our strategies, we follow Diebold and Yilmaz (2009) and construct an alternative volatility measure using daily price ranges from intraday data. Specifically, we calculate daily-range-based volatility as follows:

$$\sigma_{it}^2 = 0.511(H_{it} - L_{it})^2 - 0.019[(C_{it} - O_{it})(H_{it} + L_{it} - 2O_{it}) \\ - 2(H_{it} - O_{it})(L_{it} - O_{it})] - 0.383(C_{it} - O_{it})^2$$

where H_{it}, L_{it}, O_{it} and C_{it} denote log values of daily high, low, opening, and closing prices of commodity i on day t, respectively.

Since our continuous volatility measure relies explicitly on high-frequency intraday returns rather than daily price ranges, this alternative volatility measure only applies to realized volatility. Table 11 presents the portfolio performance of our realized volatility-connectedness strategies using this alternative measure. Results consistently indicate that these strategies achieve higher Sharpe ratios and winning ratios compared to the Markowitz benchmark, and also effectively manage tail risks. Moreover, at the monthly rebalancing frequency, our strategies deliver superior cumulative returns relative to the benchmark. These findings reinforce our main conclusions and confirm their robustness to the choice of realized volatility specifications.

6.4. *Sensitivity analysis of key parameters*

We perform sensitivity analyses to examine the robustness of our volatility-connectedness strategies across various key parameter settings, including forgetting factors, lag lengths, forecast horizons, and the scaling parameter in the penalty term. In sum, these sensitivity analyses confirm the robustness and reliability of our main findings across various model specifications and parameter choices.

First, our initial choice of forgetting factors (0.99, 0.96) follows standard practices widely accepted in the existing literature (e.g. Koop and Korobilis 2013; Antonakakis *et al.* 2020), where the first parameter ($k_1 = 0.99$) is typically set close to unity to allow gradual and stable evolution of regression parameters, and the second parameter ($k_2 = 0.96$) slightly lower to capture more rapid adjustments in volatility dynamics. Given that a setting of $k_1 = 0.99$ for parameter dynamics is well-established in the literature, we maintained this standard value fixed and conducted sensitivity analyses by varying k_2, which controls volatility dynamics. Specifically, we tested k_2 values of 0.97, 0.98, and 0.99. As shown in tables B1–B3 in

Table 7. Portfolio performance of connectedness strategy: non-crisis periods (March 2018–December 2019).

Panel A: Portfolio performance of connectedness strategy (RV)					Panel B: Portfolio performance of connectedness strategy (CV)				
	Holding period: d = 5					Holding period: d = 5			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	3.88%	5.89%	0.65	52.17%	EW	3.88%	5.89%	0.65	52.17%
Markowitz	46.37%	21.60%	1.09	63.04%	Markowitz	46.37%	21.60%	1.09	63.04%
MCoP1	7.49%	44.60%	0.16	52.17%	MCoP1	20.17%	40.45%	0.42	56.52%
MCoP2	5.56%	23.41%	0.26	52.17%	MCoP2	4.03%	29.22%	0.16	54.35%
PCI-Adjusted	− 19.06%	23.50%	− 0.71	41.30%	PCI-Adjusted	− 19.80%	23.56%	− 0.73	41.30%
NC-Adjusted	17.58%	24.31%	0.37	58.70%	NC-Adjusted	37.72%	24.14%	0.89	58.70%
NC-Restricted	− 9.59%	16.07%	− 0.37	41.30%	NC-Restricted	− 5.40%	24.98%	− 0.26	52.17%
DC-Restricted	10.01%	17.93%	0.35	43.48%	DC-Restricted	31.98%	3.91%	2.12	58.70%
	Holding period: d = 20					Holding period: d = 20			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	5.21%	4.36%	0.73	63.64%	EW	5.21%	4.36%	0.73	63.64%
Markowitz	− 2.18%	21.16%	− 0.05	45.45%	Markowitz	− 2.18%	21.16%	− 0.05	45.45%
MCoP1	− 28.92%	53.62%	− 0.62	45.45%	MCoP1	− 24.08%	52.90%	− 0.47	45.45%
MCoP2	− 7.99%	26.09%	− 0.38	54.55%	MCoP2	− 15.67%	36.03%	− 0.51	54.55%
PCI-Adjusted	− 25.05%	15.92%	− 0.93	54.55%	PCI-Adjusted	− 25.38%	16.30%	− 0.95	54.55%
NC-Adjusted	− 29.31%	24.34%	− 0.56	36.36%	NC-Adjusted	− 7.73%	28.33%	− 0.17	45.45%
NC-Restricted	− 23.66%	13.26%	− 0.89	54.55%	NC-Restricted	0.11%	5.81%	0.01	54.55%
DC-Restricted	− 4.28%	16.88%	− 0.24	45.45%	DC-Restricted	9.44%	4.01%	1.16	63.64%

Note: This table presents the portfolio performance of our proposed strategies during the non-crisis period. Panel A reports the results of strategies based on realized volatility (RV) connectedness, while Panel B reports the results of strategies based on continuous volatility (CV) connectedness. ACC_RET refers to the accumulated returns, while MAX_DD represents the maximum drawdown. The Sharpe Ratio is calculated as the excess weekly (or monthly) returns divided by the standard deviation of returns. The Winning Ratio is the number of positive weekly (or monthly) returns divided by the total number of weekly (or monthly) returns. EW refers to the equal-weighted portfolio, while Markowitz denotes the tangency portfolio that maximizes the Sharpe ratio. MCoP1 and MCoP2 represent the minimum connectedness portfolios obtained through the PCI and NPDC strategies, respectively. PCI-Adjusted and NC-Adjusted refer to the portfolios obtained through the 'PCI-Adjusted' and 'NC-Adjusted' strategies, respectively. NC-Restricted and DC-Restricted denote to the portfolios obtained through the 'NC-Restricted' and 'DC-Restricted' strategies, respectively.

Appendix B, all proposed strategies, except the 'NC-adjusted' strategy, consistently outperform the Markowitz benchmark across these settings, thereby confirming the robustness of our main conclusions.

Second, to examine the sensitivity of our results to the VAR model's lag length, we apply various lag selection criteria (AIC, HQIC, BIC, and FPE). While our benchmark analysis relies on the AIC criterion for its robustness and fewer distributional assumptions, and a lag length of 2 is used; alternative criteria (HQIC and BIC) suggest one-period lag for realized volatility, and BIC alone selects a one-period lag for continuous volatility. We thus re-estimate the TVP-VAR model using a lag length of one and recalculate the corresponding volatility connectedness measures. Table B4 in Appendix B shows that under this alternative lag specification, our connectedness-based strategies continue to outperform the Markowitz benchmark, achieving higher Sharpe ratios, higher winning ratios, and better downside risk control. These results reinforce the robustness of our findings to alternative lag-length choices.

Third, we extend the forecast horizon beyond the initial 10 periods, specifically to H = 12 and H = 14, to examine how longer horizons influence volatility connectedness measurement and portfolio performance. A longer forecast horizon may allow for capturing more persistent risk transmissions and enhanced information content. Tables B5 and B6 indicate that our strategies consistently outperform Markowitz, exhibiting higher Sharpe ratios, better winning ratios, and lower maximum drawdowns. Notably, after filtering jumps, continuous volatility-based strategies consistently outperform those using realized volatility, particularly at monthly rebalancing frequencies.

Finally, we evaluate the sensitivity of the 'PCI-Adjusted' strategy† to variations in the parameter α within its penalty term of equation (16). A larger α places greater emphasis on the PCI measure, while setting $\alpha = 0$ corresponds exactly to the standard Markowitz approach. Beyond the baseline value (10^{-5}), Table B7 presents results for alternative scaling parameters, i.e.$\alpha \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$. Our connectedness-based strategies uniformly deliver higher Sharpe ratios, improved winning ratios, and lower maximum drawdowns compared to the Markowitz benchmark across these specifications, confirming the robustness of our main findings. Furthermore, incorporating volatility connectedness information, i.e. $\alpha > 0$, steadily enhances portfolio performance relative to the Markowitz approach, demonstrating clear value-added from explicitly integrating connectedness measures into portfolio construction.

† We focus solely on the 'PCI-Adjusted' strategy for this sensitivity analysis because it has demonstrated robust and superior performance in previous tests, whereas the 'NC-Adjusted' strategy generally underperformed, with only occasional exceptions, making further sensitivity analysis on its scaling parameter (β) unnecessary.

Table 8. Portfolio performance of connectedness strategy: crisis periods (January 2020–April 2022).

Panel A: Portfolio performance of connectedness strategy (RV)					Panel B: Portfolio performance of connectedness strategy (CV)				
	Holding period: d = 5					Holding period: d = 5			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	51.18%	11.11%	2.28	64.79%	EW	51.18%	11.11%	2.28	64.79%
Markowitz	103.96%	66.84%	0.99	59.15%	Markowitz	103.96%	66.84%	0.99	59.15%
MCoP1	98.41%	57.01%	0.88	57.75%	MCoP1	84.74%	62.41%	0.79	59.15%
MCoP2	59.03%	51.98%	0.66	54.93%	MCoP2	67.17%	46.19%	0.85	59.15%
PCI-Adjusted	48.83%	13.49%	1.82	59.15%	PCI-Adjusted	41.93%	11.11%	1.95	60.56%
NC-Adjusted	49.92%	69.58%	0.50	61.97%	NC-Adjusted	137.95%	51.90%	1.42	57.75%
NC-Restricted	47.47%	13.61%	1.79	60.56%	NC-Restricted	42.45%	11.11%	1.97	60.56%
DC-Restricted	58.49%	11.11%	2.16	63.38%	DC-Restricted	44.74%	13.91%	1.82	57.75%
	Holding period: d = 20					Holding period: d = 20			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	47.24%	4.93%	2.82	64.71%	EW	47.24%	4.93%	2.82	64.71%
Markowitz	84.07%	60.03%	0.70	64.71%	Markowitz	84.07%	60.03%	0.70	64.71%
MCoP1	25.51%	52.61%	0.25	47.06%	MCoP1	18.97%	58.34%	0.17	41.18%
MCoP2	− 11.88%	53.54%	− 0.15	41.18%	MCoP2	24.78%	42.87%	0.35	58.82%
PCI-Adjusted	46.28%	6.40%	2.56	76.47%	PCI-Adjusted	38.92%	4.93%	2.54	64.71%
NC-Adjusted	− 0.76%	67.37%	− 0.01	58.82%	NC-Adjusted	74.13%	48.62%	0.79	58.82%
NC-Restricted	45.38%	6.40%	2.54	76.47%	NC-Restricted	39.28%	4.93%	2.55	64.71%
DC-Restricted	42.31%	4.93%	2.75	64.71%	DC-Restricted	45.22%	6.67%	2.06	64.71%

Note: This table presents the portfolio performance of our proposed strategies during the crisis period. Panel A reports the results of strategies based on realized volatility (RV) connectedness, while Panel B reports the results of strategies based on continuous volatility (CV) connectedness. ACC_RET refers to the accumulated returns, while MAX_DD represents the maximum drawdown. The Sharpe Ratio is calculated as the excess weekly (or monthly) returns divided by the standard deviation of returns. The Winning Ratio is the number of positive weekly (or monthly) returns divided by the total number of weekly (or monthly) returns. EW refers to the equal-weighted portfolio, while Markowitz denotes the tangency portfolio that maximizes the Sharpe ratio. MCoP1 and MCoP2 represent the minimum connectedness portfolios obtained through the PCI and NPDC strategies, respectively. PCI-Adjusted and NC-Adjusted refer to the portfolios obtained through the 'PCI-Adjusted' and 'NC-Adjusted' strategies, respectively. NC-Restricted and DC-Restricted denote to the portfolios obtained through the 'NC-Restricted' and 'DC-Restricted' strategies, respectively.

7. Conclusion

In this paper, we use 15-minute high-frequency intraday data of 17 futures from China's commodity market, covering the period from April 2019 to April 2022. We apply the Minimum Realized Variance (MinRV) estimator, as proposed by Andersen *et al.* (2012), to identify intraday jumps. By filtering out these jumps from realized volatility, we obtain the continuous volatility. We then employ the Time-Varying Parameter Vector Autoregression (TVP-VAR) model and generalized variance decomposition to construct dynamic connectedness for both overall volatility and its continuous component. Continuous volatility connectedness is considered a more effective proxy for systemic risks, as its spikes closely align with extreme risk events in the time series.

Furthermore, we apply dynamic connectedness to enhance the Markowitz portfolio strategy through two approaches: (1) revising the objective functions, and (2) imposing additional constraints. To modify the objective functions, we follow Broadstock *et al.* (2022), deriving the PCI (Pairwise Connectedness Index) from the connectedness network and developing a 'PCI-Adjusted' strategy by incorporating a PCI-penalty term into the covariance matrix. Additionally, inspired by Pedersen *et al.* (2021), we propose an 'NC (Net Connectedness)-adjusted' strategy, which directly adds an NC-penalty term to the Sharpe ratio. Regarding additional constraints, we present both 'NC-Restricted' and 'DC (Degree Centrality)-Restricted' strategies, based on net connectedness and degree centrality, respectively. These strategies assign lower weights to commodities with higher network importance, as supported by the theoretical framework of Peralta and Zareei (2016). Our portfolios are rebalanced on a weekly and monthly basis, with investment performance evaluated using four key indicators: accumulated returns, maximum drawdown, Sharpe ratio, and winning ratio. Our findings demonstrate that the proposed strategies significantly outperform the benchmark portfolios. Moreover, the continuous volatility-based strategies outperform their realized volatility counterparts in most cases, offering enhanced stability and reduced short positions. These findings are robust to alternative volatility measure, different lag lengths, extended forecast horizons, and various adjustments to key model parameters. Moreover, the superior performances of our proposed strategies are much stronger during crisis than non-turbulent periods. This suggests that volatility connectedness provides valuable incremental information for portfolio construction, especially when there are structural changes in risk transmission network due to extreme events.

While volatility connectedness effectively captures risk spillovers among assets, it does not directly measure asset return co-movements, which are explicitly represented by covariance matrices. Furthermore, estimating

Table 9. Bootstrap significance of connectedness strategy: full sample (April 2019–April 2022).

Panel A: Bootstrap significance of connectedness strategy (RV)			Panel B: Bootstrap significance of connectedness strategy (CV)		
	Holding period: d = 5			Holding period: d = 5	
	Mean Sharpe Ratio	Excess Sharpe Ratio		Mean Sharpe Ratio	Excess Sharpe Ratio
PCI-Adjusted	1.26	0.66	PCI-Adjusted	1.23*	0.65
NC-Adjusted	0.70	0.13	NC-Adjusted	0.05	– 0.54
NC-Restricted	1.30	0.69	NC-Restricted	1.25*	0.66
DC-Restricted	1.27*	0.69	DC-Restricted	1.08	0.51
	Holding period: d = 20			Holding period: d = 20	
	Mean Sharpe Ratio	Excess Sharpe Ratio		Mean Sharpe Ratio	Excess Sharpe Ratio
PCI-Adjusted	2.13**	1.56**	PCI-Adjusted	2.10**	1.53*
NC-Adjusted	0.93	0.36	NC-Adjusted	0.96	0.40
NC-Restricted	2.10**	1.54**	NC-Restricted	2.15**	1.58*
DC-Restricted	1.32	0.74	DC-Restricted	2.22**	1.64*

Note: This table presents the bootstrap significance of Sharpe ratio for our proposed strategies during full sample period. By sampling from the original return series with replacement for 5,000 times, we calculate the mean Sharpe ratio for each strategy, reported in column 'Mean Sharpe Ratio', and obtain the difference in Sharpe Ratio between our strategies and Markowitz, as is shown in column 'Excess Sharpe Ratio'. Panel A reports the results of strategies based on realized volatility (RV) connectedness, while Panel B reports the results of strategies based on continuous volatility (CV) connectedness. PCI-Adjusted and NC-Adjusted refer to the portfolios obtained through the 'PCI-Adjusted' and 'NC-Adjusted' strategies, respectively. NC-Restricted and DC-Restricted denote to the portfolios obtained through the 'NC-Restricted' and 'DC-Restricted' strategies, respectively. Null hypothesis includes: (i) Mean Sharpe Ratio ≤ 0; (ii) Excess Sharpe Ratio ≤ 0. The asterisks ($***$, $**$, $*$) indicate rejection of the hypothesis at the 1%, 5%, 10% significance level, respectively.

Table 10. Bootstrap significance of connectedness strategy: crisis period (January 2020–April 2022).

Panel A: Bootstrap significance of connectedness strategy (RV)			Panel B: Bootstrap significance of connectedness strategy (CV)		
	Holding period: d = 5			Holding period: d = 5	
	Mean Sharpe Ratio	Excess Sharpe Ratio		Mean Sharpe Ratio	Excess Sharpe Ratio
PCI-Adjusted	1.87**	0.84	PCI-Adjusted	1.98**	0.95
NC-Adjusted	0.52	– 0.50	NC-Adjusted	1.45*	0.41
NC-Restricted	1.84**	0.81	NC-Restricted	2.00**	0.97
DC-Restricted	2.20***	1.17*	DC-Restricted	1.86**	0.83
	Holding period: d = 20			Holding period: d = 20	
	Mean Sharpe Ratio	Excess Sharpe Ratio		Mean Sharpe Ratio	Excess Sharpe Ratio
PCI-Adjusted	2.80***	1.97***	PCI-Adjusted	2.66***	1.86**
NC-Adjusted	– 0.03	– 0.83	NC-Adjusted	0.89	0.09
NC-Restricted	2.79***	1.96**	NC-Restricted	2.67***	1.87**
DC-Restricted	2.88***	2.09***	DC-Restricted	2.14***	1.35**

Note: This table presents the bootstrap significance of Sharpe ratio for our proposed strategies during the crisis period. By sampling from the original return series with replacement for 5,000 times, we calculate the mean Sharpe ratio for each strategy, reported in column 'Mean Sharpe Ratio', and obtain the difference in Sharpe Ratio between our strategies and Markowitz, as is shown in column 'Excess Sharpe Ratio'. Panel A reports the results of strategies based on realized volatility (RV) connectedness, while Panel B reports the results of strategies based on continuous volatility (CV) connectedness. PCI-Adjusted and NC-Adjusted refer to the portfolios obtained through the 'PCI-Adjusted' and 'NC-Adjusted' strategies, respectively. NC-Restricted and DC-Restricted denote to the portfolios obtained through the 'NC-Restricted' and 'DC-Restricted' strategies, respectively. Null hypothesis includes: (i) Mean Sharpe Ratio ≤ 0; (ii Excess Sharpe Ratio ≤ 0. The asterisks ($***$, $**$, $*$) indicate rejection of the hypothesis at the 1%, 5%, 10% significance level, respectively.

volatility connectedness involves model parameters, potentially introducing additional estimation uncertainty. Consequently, completely discarding covariance information (as in MCoP1/MCoP2) may lead to weaker portfolio performance. The complementary roles of covariance and connectedness are further confirmed by our sensitivity analysis on the PCI-Adjusted strategy, where we vary the scaling parameter in its penalty term. Results clearly indicate that strategies integrating both covariance and connectedness consistently outperform the pure covariance-based (Markowitz) strategy. Thus, incorporating volatility connectedness alongside covariance significantly enhances portfolio optimization outcomes.

In summary, our research introduces new methods for portfolio optimization based on volatility connectedness. These strategies significantly outperform the traditional Markowitz approach and other benchmark models in most cases, highlighting the value of incorporating volatility connectedness in portfolio construction. Moreover, continuous volatility connectedness provides a more accurate representation of systemic risks and enhances investment performance. This

Table 11. Portfolio performance of connectedness strategy: using logarithmic volatilities.

Panel A: Portfolio performance of connectedness strategy				
	Holding period: d = 5			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	51.17%	12.61%	1.91	57.69%
Markowitz	53.24%	64.14%	0.56	51.92%
MCoP1	− 5.95%	68.69%	− 0.06	48.08%
MCoP2	3.55%	60.92%	0.04	50.96%
PCI-Adjusted	43.74%	13.90%	1.35	62.50%
NC-Adjusted	− 14.33%	47.17%	− 0.19	51.92%
NC-Restricted	45.77%	13.96%	1.42	60.58%
DC-Restricted	47.05%	16.03%	1.25	59.62%
	Holding period: d = 20			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	61.76%	9.04%	2.49	84.62%
Markowitz	46.63%	60.58%	0.46	65.38%
MCoP1	69.10%	62.09%	0.47	65.38%
MCoP2	30.68%	51.82%	0.33	57.69%
PCI-Adjusted	49.71%	16.50%	1.56	76.92%
NC-Adjusted	63.49%	33.30%	0.60	61.54%
NC-Restricted	50.18%	16.40%	1.61	80.77%
DC-Restricted	74.71%	9.04%	2.33	84.62%

Note: This table presents the portfolio performance via using intraday realized volatility measure proposed by Diebold and Yilmaz (2009). ACC_RET refers to the accumulated returns, while MAX_DD represents the maximum drawdown. The Sharpe Ratio is calculated as the excess weekly (or monthly) returns divided by the standard deviation of returns. The Winning Ratio is the number of positive weekly (or monthly) returns divided by the total number of weekly (or monthly) returns. EW refers to the equal-weighted portfolio, while Markowitz denotes the tangency portfolio that maximizes the Sharpe ratio. MCoP1 and MCoP2 represent the minimum connectedness portfolios obtained through the PCI and NPDC strategies, respectively. PCI-Adjusted and NC-Adjusted refer to the portfolios obtained through the 'PCI-Adjusted' and 'NC-Adjusted' strategies, respectively.

improvement may result from the distortion caused by intraday jumps, which obscure connectedness effects.

Our findings offer several important practical implications beyond methodological contributions. For regulators, the volatility connectedness measures proposed here can serve as real-time early-warning tools for monitoring systemic risk across commodity markets, identifying critical nodes of risk transmission, and guiding timely regulatory responses, including adjustments in margin requirements or coordinated market interventions. For institutional investors and portfolio managers, incorporating dynamic volatility connectedness into portfolio decisions can significantly enhance their risk management capabilities and adaptability, particularly in periods of market turbulence. These insights highlight the practical value of adopting network-based approaches, contributing to both academic research and effective market surveillance.

While our findings demonstrate the effectiveness of network-based portfolio strategies from a theoretical and empirical perspective, it is important to acknowledge several practical considerations regarding their implementation in real-world settings. First, transaction costs and liquidity constraints may affect the net performance of the strategies, particularly in less actively traded commodity futures. Although the current analysis does not explicitly incorporate transaction costs, the use of restricted-weight portfolios and the relatively stable weight dynamics observed in our strategies may help limit turnover and mitigate excessive trading costs. Second, the applicability of our framework in other market contexts may depend on institutional and structural differences, such as trading frequency, market depth, and regulatory regimes. Our study focuses specifically on China's commodity futures market, characterized by frequent regulatory interventions, dominance by retail investors, and synchronized exchange-level supervision. These institutional features may amplify volatility spillovers during market stress periods, emphasizing the particular effectiveness of our network-based strategies in conditions of elevated systemic risk. Future research could test the generalizability of this framework in different markets, while also incorporating explicit transaction costs and liquidity constraints to enhance its practical relevance.

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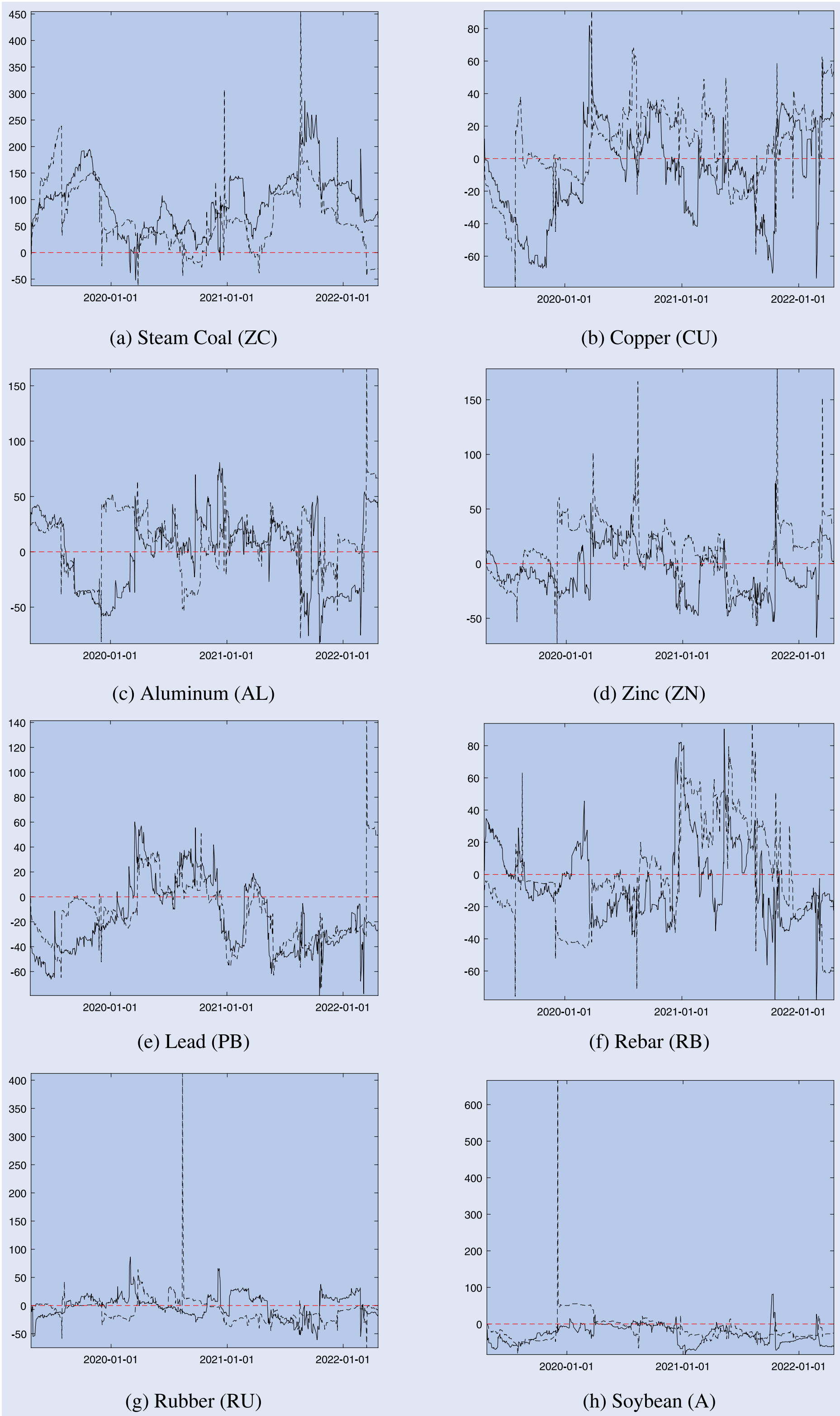
Appendices

Appendix A



(a) Steam Coal (ZC)

(b) Copper (CU)

(c) Aluminum (AL)

(d) Zinc (ZN)

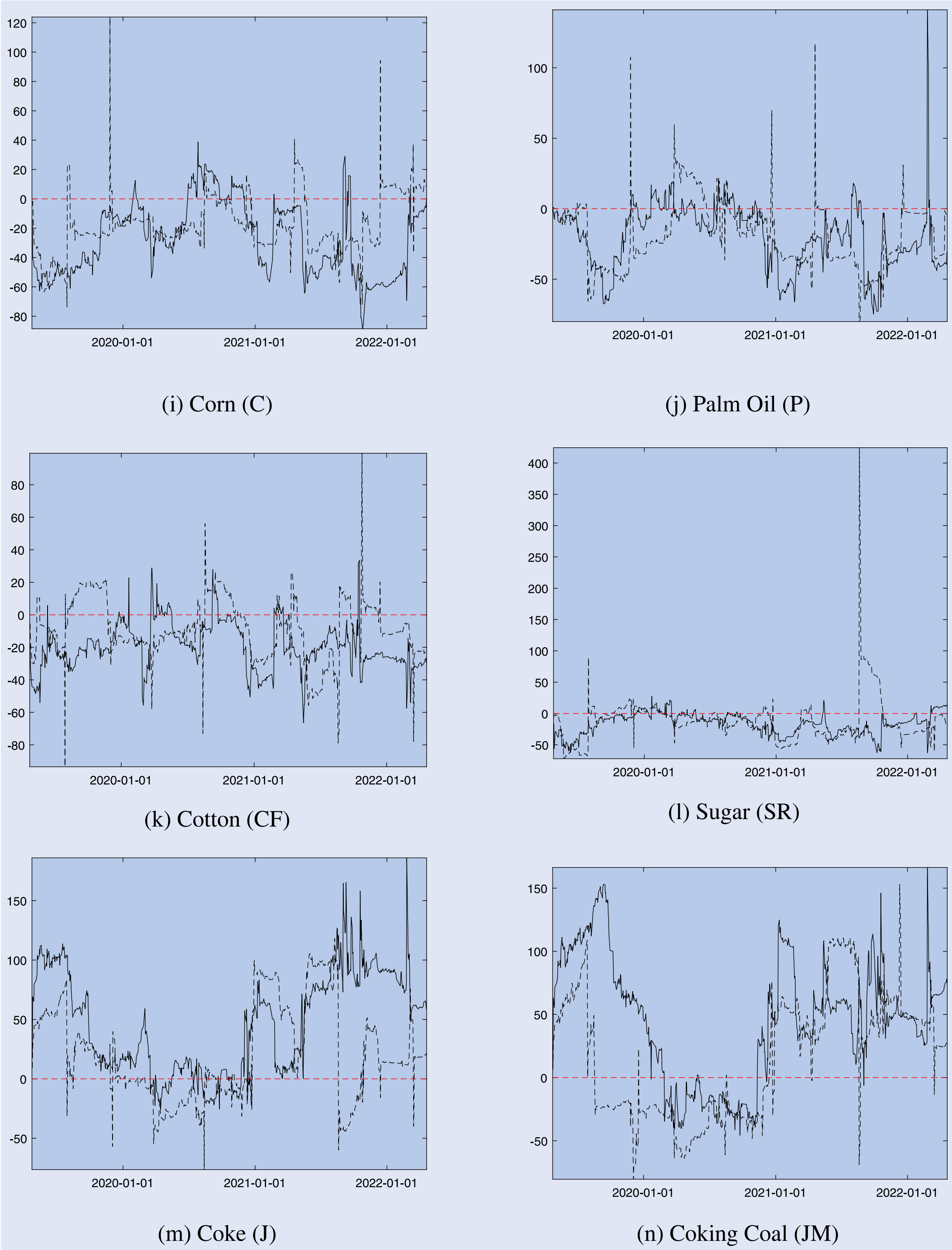
(e) Lead (PB)

(f) Rebar (RB)

(g) Rubber (RU)

(h) Soybean (A)

Figure A1. Net connectedness: realized volatility and continuous volatility.
Note: This figure illustrates the dynamic net connectedness of realized and continuous volatility, estimated using the TVP-VAR method. The vertical axis denotes net connectedness, and the horizontal axis indicates the date. The dashed line represents the net connectedness of realized volatility, while the solid line shows the net connectedness of continuous volatility. The parameters are set as follows: lag order $p = 2$; forecast horizon $H = 10$; forgetting factor $(k_1, k_2) = (0.99, 0.96)$.



(i) Corn (C)

(j) Palm Oil (P)

(k) Cotton (CF)

(l) Sugar (SR)

(m) Coke (J)

(n) Coking Coal (JM)

Figure A1. Continued.

Appendix B

Table B1. Portfolio performance of connectedness strategy: alternative forgetting factors ($\boldsymbol{k}_1 = 0.99, \boldsymbol{k}_2 = 0.97$).

Panel A: Portfolio performance of connectedness strategy (RV)					Panel B: Portfolio performance of connectedness strategy (CV)				
	Holding period: d = 5					Holding period: d = 5			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	51.17%	12.61%	1.91	57.69%	EW	51.17%	12.61%	1.91	57.69%
Markowitz	53.24%	64.14%	0.56	51.92%	Markowitz	53.24%	64.14%	0.56	51.92%
MCoP1	5.49%	63.75%	0.05	51.92%	MCoP1	16.19%	64.41%	0.17	49.04%
MCoP2	41.15%	46.58%	0.48	53.85%	MCoP2	31.63%	57.91%	0.37	52.88%
PCI-Adjusted	49.65%	14.23%	1.31	59.62%	PCI-Adjusted	67.03%	11.10%	1.80	55.77%
NC-Adjusted	59.74%	56.37%	0.58	54.81%	NC-Adjusted	32.71%	41.96%	0.36	51.92%
NC-Restricted	48.32%	14.43%	1.25	58.65%	NC-Restricted	68.32%	11.10%	1.82	55.77%
DC-Restricted	72.89%	13.86%	1.53	59.62%	DC-Restricted	43.54%	20.72%	0.94	59.62%
	Holding period: d = 20					Holding period: d = 20			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	61.76%	9.04%	2.49	84.62%	EW	61.76%	9.04%	2.49	84.62%
Markowitz	46.63%	60.58%	0.46	65.38%	Markowitz	46.63%	60.58%	0.46	65.38%
MCoP1	64.00%	56.93%	0.48	61.54%	MCoP1	75.57%	57.43%	0.56	57.69%
MCoP2	89.03%	38.94%	0.88	61.54%	MCoP2	32.70%	49.06%	0.38	57.69%
PCI-Adjusted	47.93%	16.90%	1.44	73.08%	PCI-Adjusted	86.24%	8.44%	2.46	80.77%
NC-Adjusted	93.21%	51.89%	0.70	61.54%	NC-Adjusted	57.57%	35.90%	0.58	65.38%
NC-Restricted	49.90%	17.23%	1.45	76.92%	NC-Restricted	85.98%	8.45%	2.46	80.77%
DC-Restricted	49.81%	8.70%	2.07	65.38%	DC-Restricted	82.39%	9.04%	2.36	84.62%

Note: This table presents the portfolio performance of our proposed strategies with alternative forgetting factors ($k_1 = 0.99, k_2 = 0.97$). Panel A reports the results of strategies based on realized volatility (RV) connectedness, while Panel B reports the results of strategies based on continuous volatility (CV) connectedness. ACC_RET refers to the accumulated returns, while MAX_DD represents the maximum drawdown. The Sharpe Ratio is calculated as the excess weekly (or monthly) returns divided by the standard deviation of returns. The Winning Ratio is the number of positive weekly (or monthly) returns divided by the total number of weekly (or monthly) returns. EW refers to the equal-weighted portfolio, while Markowitz denotes the tangency portfolio that maximizes the Sharpe ratio. MCoP1 and MCoP2 represent the minimum connectedness portfolios obtained through the PCI and NPDC strategies, respectively. PCI-Adjusted and NC-Adjusted refer to the portfolios obtained through the 'PCI-Adjusted' and 'NC-Adjusted' strategies, respectively. NC-Restricted and DC-Restricted denote to the portfolios obtained through the 'NC-Restricted' and 'DC-Restricted' strategies, respectively.

Table B2. Portfolio performance of connectedness strategy: alternative forgetting factors ($\boldsymbol{k}_1 = 0.99, \boldsymbol{k}_2 = 0.98$).

Panel A: Portfolio performance of connectedness strategy (RV)					Panel B: Portfolio performance of connectedness strategy (CV)				
	Holding period: d = 5					Holding period: d = 5			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	51.17%	12.61%	1.91	57.69%	EW	51.17%	12.61%	1.91	57.69%
Markowitz	53.24%	64.14%	0.56	51.92%	Markowitz	53.24%	64.14%	0.56	51.92%
MCoP1	16.95%	65.06%	0.17	54.81%	MCoP1	48.06%	62.37%	0.47	50.96%
MCoP2	44.83%	55.71%	0.46	56.73%	MCoP2	38.46%	62.19%	0.40	55.77%
PCI-Adjusted	49.55%	12.61%	1.35	54.81%	PCI-Adjusted	39.71%	22.12%	1.06	53.85%
NC-Adjusted	45.03%	70.14%	0.35	51.92%	NC-Adjusted	− 15.55%	48.32%	− 0.19	50.96%
NC-Restricted	50.27%	12.61%	1.32	54.81%	NC-Restricted	39.36%	23.03%	1.04	53.85%
DC-Restricted	76.85%	11.93%	1.68	55.77%	DC-Restricted	45.40%	15.39%	1.35	59.62%
	Holding period: d = 20					Holding period: d = 20			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	61.76%	9.04%	2.49	84.62%	EW	61.76%	9.04%	2.49	84.62%
Markowitz	46.63%	60.58%	0.46	65.38%	Markowitz	46.63%	60.58%	0.46	65.38%
MCoP1	71.24%	58.60%	0.53	61.54%	MCoP1	93.78%	55.38%	0.69	61.54%
MCoP2	55.84%	49.57%	0.58	65.38%	MCoP2	35.53%	54.51%	0.38	61.54%
PCI-Adjusted	59.42%	9.04%	1.90	76.92%	PCI-Adjusted	72.63%	11.12%	2.29	80.77%
NC-Adjusted	68.93%	63.48%	0.46	61.54%	NC-Adjusted	38.18%	41.60%	0.36	69.23%
NC-Restricted	62.94%	9.04%	1.95	80.77%	NC-Restricted	71.93%	11.27%	2.34	80.77%
DC-Restricted	46.46%	8.70%	1.77	65.38%	DC-Restricted	63.44%	6.48%	2.68	80.77%

Note: This table presents the portfolio performance of our proposed strategies with alternative forgetting factors ($k_1 = 0.99, k_2 = 0.98$). Panel A reports the results of strategies based on realized volatility (RV) connectedness, while Panel B reports the results of strategies based on continuous volatility (CV) connectedness. ACC_RET refers to the accumulated returns, while MAX_DD represents the maximum drawdown. The Sharpe Ratio is calculated as the excess weekly (or monthly) returns divided by the standard deviation of returns. The Winning Ratio is the number of positive weekly (or monthly) returns divided by the total number of weekly (or monthly) returns. EW refers to the equal-weighted portfolio, while Markowitz denotes the tangency portfolio that maximizes the Sharpe ratio. MCoP1 and MCoP2 represent the minimum connectedness portfolios obtained through the PCI and NPDC strategies, respectively. PCI-Adjusted and NC-Adjusted refer to the portfolios obtained through the 'PCI-Adjusted' and 'NC-Adjusted' strategies, respectively. NC-Restricted and DC-Restricted denote to the portfolios obtained through the 'NC-Restricted' and 'DC-Restricted' strategies, respectively.

Table B3. Portfolio performance of connectedness strategy: alternative forgetting factors ($\boldsymbol{k}_1 = 0.99, \boldsymbol{k}_2 = 0.99$).

	Panel A: Portfolio performance of connectedness strategy (RV)					Panel B: Portfolio performance of connectedness strategy (CV)			
	Holding period: d = 5					Holding period: d = 5			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	51.17%	12.61%	1.91	57.69%	EW	51.17%	12.61%	1.91	57.69%
Markowitz	53.24%	64.14%	0.56	51.92%	Markowitz	53.24%	64.14%	0.56	51.92%
MCoP1	52.05%	67.01%	0.47	62.50%	MCoP1	53.95%	61.94%	0.52	53.85%
MCoP2	21.29%	73.83%	0.19	54.81%	MCoP2	35.95%	66.31%	0.33	55.77%
PCI-Adjusted	51.65%	14.30%	1.37	53.85%	PCI-Adjusted	53.91%	15.34%	1.52	58.65%
NC-Adjusted	− 12.22%	70.70%	− 0.12	52.88%	NC-Adjusted	− 38.94%	64.92%	− 0.42	51.92%
NC-Restricted	50.97%	15.01%	1.32	53.85%	NC-Restricted	53.44%	15.84%	1.56	60.58%
DC-Restricted	64.52%	11.47%	1.60	54.81%	DC-Restricted	38.08%	15.87%	1.21	59.62%
	Holding period: d = 20					Holding period: d = 20			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	61.76%	9.04%	2.49	84.62%	EW	61.76%	9.04%	2.49	84.62%
Markowitz	46.63%	60.58%	0.46	65.38%	Markowitz	46.63%	60.58%	0.46	65.38%
MCoP1	74.81%	60.48%	0.61	61.54%	MCoP1	76.92%	55.55%	0.60	61.54%
MCoP2	51.87%	67.35%	0.42	73.08%	MCoP2	66.36%	59.74%	0.54	61.54%
PCI-Adjusted	57.70%	7.91%	1.84	73.08%	PCI-Adjusted	65.87%	11.10%	2.19	80.77%
NC-Adjusted	21.92%	65.01%	0.17	50.00%	NC-Adjusted	66.20%	62.43%	0.41	65.38%
NC-Restricted	53.70%	7.84%	1.65	69.23%	NC-Restricted	63.57%	11.27%	2.27	80.77%
DC-Restricted	83.05%	7.84%	3.17	76.92%	DC-Restricted	51.39%	8.46%	2.19	73.08%

Note: This table presents the portfolio performance of our proposed strategies with alternative forgetting factors ($k_1 = 0.99, k_2 = 0.99$). Panel A reports the results of strategies based on realized volatility (RV) connectedness, while Panel B reports the results of strategies based on continuous volatility (CV) connectedness. ACC_RET refers to the accumulated returns, while MAX_DD represents the maximum drawdown. The Sharpe Ratio is calculated as the excess weekly (or monthly) returns divided by the standard deviation of returns. The Winning Ratio is the number of positive weekly (or monthly) returns divided by the total number of weekly (or monthly) returns. EW refers to the equal-weighted portfolio, while Markowitz denotes the tangency portfolio that maximizes the Sharpe ratio. MCoP1 and MCoP2 represent the minimum connectedness portfolios obtained through the PCI and NPDC strategies, respectively. PCI-Adjusted and NC-Adjusted refer to the portfolios obtained through the 'PCI-Adjusted' and 'NC-Adjusted' strategies, respectively. NC-Restricted and DC-Restricted denote to the portfolios obtained through the 'NC-Restricted' and 'DC-Restricted' strategies, respectively.

Table B4. Portfolio performance of connectedness strategy: alternative lag length ($p = 1$).

Panel A: Portfolio performance of connectedness strategy (RV)					Panel B: Portfolio performance of connectedness strategy (CV)				
	Holding period: d = 5					Holding period: d = 5			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	51.17%	12.61%	1.91	57.69%	EW	51.17%	12.61%	1.91	57.69%
Markowitz	53.24%	64.14%	0.56	51.92%	Markowitz	53.24%	64.14%	0.56	51.92%
MCoP1	10.53%	63.41%	0.10	50.96%	MCoP1	9.58%	66.74%	0.10	46.15%
MCoP2	26.29%	58.92%	0.31	54.81%	MCoP2	20.09%	59.98%	0.25	53.85%
PCI-Adjusted	87.09%	13.16%	2.08	63.46%	PCI-Adjusted	45.61%	12.61%	1.24	58.65%
NC-Adjusted	– 0.81%	60.90%	– 0.01	54.81%	NC-Adjusted	– 80.73%	84.45%	– 1.61	48.08%
NC-Restricted	83.60%	13.31%	1.97	62.50%	NC-Restricted	48.47%	12.61%	1.30	59.62%
DC-Restricted	24.92%	26.62%	0.59	58.65%	DC-Restricted	26.67%	21.69%	0.73	60.58%
	Holding period: d = 20					Holding period: d = 20			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	61.76%	9.04%	2.49	84.62%	EW	61.76%	9.04%	2.49	84.62%
Markowitz	46.63%	60.58%	0.46	65.38%	Markowitz	46.63%	60.58%	0.46	65.38%
MCoP1	76.50%	55.53%	0.60	61.54%	MCoP1	87.52%	59.78%	0.61	57.69%
MCoP2	43.58%	51.74%	0.46	57.69%	MCoP2	44.44%	51.91%	0.45	61.54%
PCI-Adjusted	53.43%	17.34%	1.38	73.08%	PCI-Adjusted	108.37%	9.04%	2.18	76.92%
NC-Adjusted	36.48%	45.71%	0.34	57.69%	NC-Adjusted	– 57.63%	71.62%	– 1.01	50.00%
NC-Restricted	54.77%	17.23%	1.38	73.08%	NC-Restricted	109.99%	9.04%	2.14	80.77%
DC-Restricted	31.77%	8.70%	1.27	73.08%	DC-Restricted	68.87%	9.04%	2.22	84.62%

Note: This table presents the portfolio performance of our proposed strategies with alternative lag length ($p = 1$). Panel A reports the results of strategies based on realized volatility (RV) connectedness, while Panel B reports the results of strategies based on continuous volatility (CV) connectedness. ACC_RET refers to the accumulated returns, while MAX_DD represents the maximum drawdown. The Sharpe Ratio is calculated as the excess weekly (or monthly) returns divided by the standard deviation of returns. The Winning Ratio is the number of positive weekly (or monthly) returns divided by the total number of weekly (or monthly) returns. EW refers to the equal-weighted portfolio, while Markowitz denotes the tangency portfolio that maximizes the Sharpe ratio. MCoP1 and MCoP2 represent the minimum connectedness portfolios obtained through the PCI and NPDC strategies, respectively. PCI-Adjusted and NC-Adjusted refer to the portfolios obtained through the 'PCI-Adjusted' and 'NC-Adjusted' strategies, respectively. NC-Restricted and DC-Restricted denote to the portfolios obtained through the 'NC-Restricted' and 'DC-Restricted' strategies, respectively.

Table B5. Portfolio performance of connectedness strategy: alternative forecast horizon (H = 12).

Panel A: Portfolio performance of connectedness strategy (RV)					Panel B: Portfolio performance of connectedness strategy (CV)				
	Holding period: d = 5					Holding period: d = 5			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	51.17%	12.61%	1.91	57.69%	EW	51.17%	12.61%	1.91	57.69%
Markowitz	53.24%	64.14%	0.56	51.92%	Markowitz	53.24%	64.14%	0.56	51.92%
MCoP1	− 1.85%	63.57%	− 0.02	52.88%	MCoP1	13.53%	65.94%	0.13	48.08%
MCoP2	15.49%	57.89%	0.18	49.04%	MCoP2	29.11%	57.99%	0.34	51.92%
PCI-Adjusted	55.00%	20.69%	1.30	59.62%	PCI-Adjusted	34.85%	11.10%	1.10	56.73%
NC-Adjusted	83.26%	45.78%	0.92	54.81%	NC-Adjusted	− 1.47%	49.24%	− 0.02	50.96%
NC-Restricted	58.96%	20.85%	1.34	59.62%	NC-Restricted	35.90%	11.10%	1.13	55.77%
DC-Restricted	59.09%	13.76%	1.19	58.65%	DC-Restricted	41.25%	20.72%	0.93	59.62%
	Holding period: d = 20					Holding period: d = 20			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	61.76%	9.04%	2.49	84.62%	EW	61.76%	9.04%	2.49	84.62%
Markowitz	46.63%	60.58%	0.46	65.38%	Markowitz	46.63%	60.58%	0.46	65.38%
MCoP1	62.43%	56.11%	0.48	61.54%	MCoP1	95.48%	58.12%	0.66	57.69%
MCoP2	27.17%	50.18%	0.29	57.69%	MCoP2	36.23%	49.54%	0.40	65.38%
PCI-Adjusted	67.01%	14.95%	1.90	88.46%	PCI-Adjusted	74.65%	8.45%	1.89	76.92%
NC-Adjusted	116.73%	39.65%	0.94	61.54%	NC-Adjusted	31.13%	38.34%	0.42	61.54%
NC-Restricted	67.17%	15.15%	1.89	84.62%	NC-Restricted	74.18%	8.45%	1.93	76.92%
DC-Restricted	33.22%	12.33%	1.07	73.08%	DC-Restricted	73.31%	9.04%	1.95	84.62%

Note: This table presents the portfolio performance of our proposed strategies with alternative forecast horizon ($H = 12$). Panel A reports the results of strategies based on realized volatility (RV) connectedness, while Panel B reports the results of strategies based on continuous volatility (CV) connectedness. ACC_RET refers to the accumulated returns, while MAX_DD represents the maximum drawdown. The Sharpe Ratio is calculated as the excess weekly (or monthly) returns divided by the standard deviation of returns. The Winning Ratio is the number of positive weekly (or monthly) returns divided by the total number of weekly (or monthly) returns. EW refers to the equal-weighted portfolio, while Markowitz denotes the tangency portfolio that maximizes the Sharpe ratio. MCoP1 and MCoP2 represent the minimum connectedness portfolios obtained through the PCI and NPDC strategies, respectively. PCI-Adjusted and NC-Adjusted refer to the portfolios obtained through the 'PCI-Adjusted' and 'NC-Adjusted' strategies, respectively. NC-Restricted and DC-Restricted denote to the portfolios obtained through the 'NC-Restricted' and 'DC-Restricted' strategies, respectively.

Table B6. Portfolio performance of connectedness strategy: alternative forecast horizon (H = 14).

Panel A: Portfolio performance of connectedness strategy (RV)					Panel B: Portfolio performance of connectedness strategy (CV)				
	Holding period: d = 5					Holding period: d = 5			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	51.17%	12.61%	1.91	57.69%	EW	51.17%	12.61%	1.91	57.69%
Markowitz	53.24%	64.14%	0.56	51.92%	Markowitz	53.24%	64.14%	0.56	51.92%
MCoP1	– 2.71%	63.41%	– 0.03	50.00%	MCoP1	13.27%	65.80%	0.13	49.04%
MCoP2	11.11%	57.80%	0.14	50.00%	MCoP2	29.09%	57.89%	0.34	53.85%
PCI-Adjusted	52.53%	20.69%	1.26	59.62%	PCI-Adjusted	40.81%	10.57%	1.26	57.69%
NC-Adjusted	80.75%	45.43%	0.84	53.85%	NC-Adjusted	37.87%	50.61%	0.36	53.85%
NC-Restricted	58.33%	20.85%	1.34	59.62%	NC-Restricted	42.00%	10.62%	1.30	56.73%
DC-Restricted	55.66%	16.38%	1.13	58.65%	DC-Restricted	44.95%	20.72%	0.97	59.62%
	Holding period: d = 20					Holding period: d = 20			
	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio		ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
EW	61.76%	9.04%	2.49	84.62%	EW	61.76%	9.04%	2.49	84.62%
Markowitz	46.63%	60.58%	0.46	65.38%	Markowitz	46.63%	60.58%	0.46	65.38%
MCoP1	63.15%	55.84%	0.48	61.54%	MCoP1	97.05%	57.92%	0.67	57.69%
MCoP2	23.76%	49.85%	0.27	57.69%	MCoP2	45.79%	49.23%	0.48	65.38%
PCI-Adjusted	64.01%	14.96%	1.83	88.46%	PCI-Adjusted	83.93%	7.61%	2.06	76.92%
NC-Adjusted	122.48%	39.30%	0.98	65.38%	NC-Adjusted	23.84%	42.83%	0.32	61.54%
NC-Restricted	64.71%	15.15%	1.84	84.62%	NC-Restricted	83.82%	7.65%	2.12	76.92%
DC-Restricted	33.87%	12.33%	1.08	73.08%	DC-Restricted	72.49%	9.04%	1.96	84.62%

Note: This table presents the portfolio performance of our proposed strategies with alternative forecast horizon (H = 14). Panel A reports the results of strategies based on realized volatility (RV) connectedness, while Panel B reports the results of strategies based on continuous volatility (CV) connectedness. ACC_RET refers to the accumulated returns, while MAX_DD represents the maximum drawdown. The Sharpe Ratio is calculated as the excess weekly (or monthly) returns divided by the standard deviation of returns. The Winning Ratio is the number of positive weekly (or monthly) returns divided by the total number of weekly (or monthly) returns. EW refers to the equal-weighted portfolio, while Markowitz denotes the tangency portfolio that maximizes the Sharpe ratio. MCoP1 and MCoP2 represent the minimum connectedness portfolios obtained through the PCI and NPDC strategies, respectively. PCI-Adjusted and NC-Adjusted refer to the portfolios obtained through the 'PCI-Adjusted' and 'NC-Adjusted' strategies, respectively. NC-Restricted and DC-Restricted denote to the portfolios obtained through the 'NC-Restricted' and 'DC-Restricted' strategies, respectively.

Table B7. Portfolio Performance of PCI-Adjusted Strategy: changing parameter in penalty-term.

Panel A: Portfolio performance of PCI-Adjusted strategy (RV)					Panel B: Portfolio performance of PCI-Adjusted strategy (CV)				
	Holding period: d = 5					Holding period: d = 5			
Scale	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio	Scale	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
Markowitz	53.24%	64.14%	0.56	51.92%	Markowitz	53.24%	64.14%	0.56	51.92%
10^{-4}	53.63%	13.48%	1.41	58.65%	10^{-4}	41.83%	11.14%	1.32	60.58%
10^{-3}	60.92%	10.59%	1.63	57.69%	10^{-3}	44.71%	12.86%	1.41	58.65%
10^{-2}	61.05%	10.46%	1.47	58.65%	10^{-2}	50.32%	15.95%	1.13	60.58%
10^{-1}	75.18%	10.45%	1.85	58.65%	10^{-1}	32.49%	16.10%	0.92	55.77%
1	59.38%	10.45%	1.58	57.69%	1	39.15%	13.24%	1.22	57.69%
	Holding period: d = 20					Holding period: d = 20			
Scale	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio	Scale	ACC_RET	MAX_DD	Sharpe Ratio	Winning Ratio
Markowitz	46.63%	60.58%	0.46	65.38%	Markowitz	46.63%	60.58%	0.46	65.38%
10^{-4}	61.69%	14.00%	1.91	88.46%	10^{-4}	73.71%	8.48%	1.90	76.92%
10^{-3}	53.75%	13.19%	1.55	84.62%	10^{-3}	77.98%	8.52%	2.05	73.08%
10^{-2}	63.19%	13.04%	1.98	80.77%	10^{-2}	35.39%	19.52%	0.92	69.23%
10^{-1}	70.62%	13.03%	2.16	84.62%	10^{-1}	59.23%	8.53%	1.60	65.38%
1	63.99%	13.02%	2.01	84.62%	1	80.36%	8.53%	2.00	73.08%

Note: This table presents the portfolio performance of PCI-adjusted strategy under different parameters values in its penalty term. When the scaling parameter equals to 0, PCI-adjusted strategy is identical to the Markowitz model. ACC_RET refers to the accumulated returns, while MAX_DD represents the maximum drawdown. The Sharpe Ratio is calculated as the excess weekly (or monthly) returns divided by the standard deviation of returns. The Winning Ratio is the number of positive weekly (or monthly) returns divided by the total number of weekly (or monthly) returns.